

Modelling Financial Markets

Chapter 11: Continuous-Time Models



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

How do we describe prices in continuous time?

What the models explain

- Prices are recorded daily or every few milliseconds: can one model describe them at **every** time scale at once?
- Part I: Brownian motion, Itô calculus, simulation, geometric Brownian motion (GBM), mean reversion
- Part II: estimating diffusions, change of measure, jumps, stochastic volatility, models against data, a case study, AI-assisted discovery

What you should be able to do

- apply Itô's lemma and simulate a stochastic differential equation with a known error order
- estimate GBM, Ornstein–Uhlenbeck, Merton and Heston on data, with standard errors that respect what the sample identifies
- say which feature of the data each model reproduces, and which feature none of them does
- explain why option pricing (Chapter 12) and interest-rate models use this language

Indices and Data Sources

- **VIX**: Cboe Volatility Index.
- **BET**: Bucharest Exchange Trading (index).
- **BVB**: Bursa de Valori București — the Bucharest Stock Exchange.
- **CBOE**: Chicago Board Options Exchange.
- **FRED**: Federal Reserve Economic Data.
- **DTB3**: 3-Month Treasury Bill: secondary market rate (FRED series code).

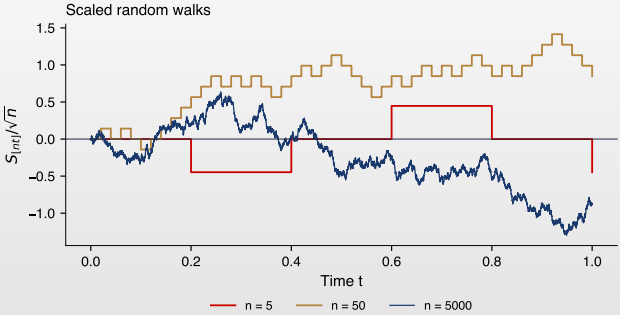
Data Used in This Chapter

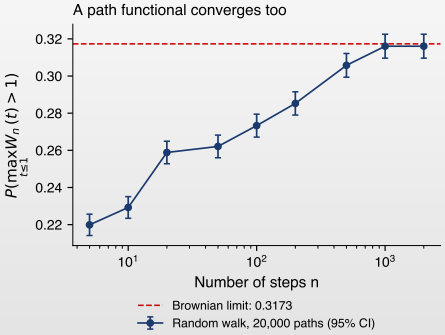
Series	Type	Period	Observations / frequency
S&P 500	US equity index	1990–2026	9 245
BET	BVB equity index	2000–2026	6 670
Bitcoin	crypto-asset, 7 days a week	2014–2026	4 384
VIX	CBOE implied volatility index of the S&P 500	1990–2026	daily
DTB3	3-month US Treasury bill rate (FRED)	1954–2026	daily

- Daily closing values up to 18 September 2026; daily log returns $r_t = \ln P_t - \ln P_{t-1}$, each series on its own calendar (cleaning rules: Appendix)
- Time in years, daily step $\Delta t = 1/252$ (Bitcoin: $1/365$); drifts and volatilities annualised
- Intraday: SPY (exchange-traded fund, ETF, tracking the S&P 500) 5-minute bars, October 2020 – September 2026, 1 468 full trading days (case study; Chapter 9); the course data contain no intraday Bitcoin series
- W_t: standard Brownian motion; S_t (also P_t): price; v_t: instantaneous variance; $\mathcal{F}_t$: the information (filtration) up to t
- $\mathrm{d}S/S$ is the instantaneous simple return and $\mathrm{d}\ln S$ the log change; Itô's lemma explains their different drifts

From Random Walk to Brownian Motion: Donsker's Theorem (1/2)

CI: Confidence Interval.





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From Random Walk to Brownian Motion: Donsker's Theorem (2/2)

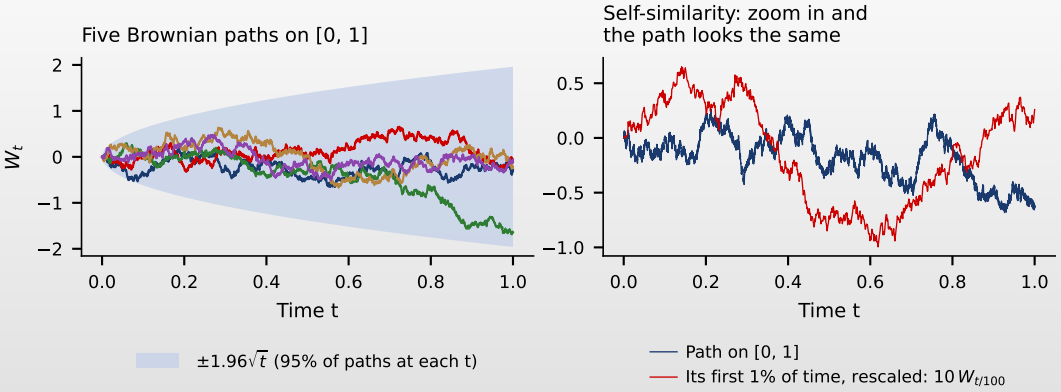
- Scaled random walk $W_n(t) = S_{\lfloor nt \rfloor}/\sqrt{n}$, $S_k = \xi_1 + \cdots + \xi_k$, steps ξ_i i.i.d. (independent and identically distributed) with mean 0 and variance 1; $\lfloor nt \rfloor$: integer part of nt; time step $1/n$, space step $1/\sqrt{n}$
  - Central limit theorem (CLT): $W_n(t) \to N(0, t)$ for each t; Donsker: $W_n \Rightarrow W$ ($\Rightarrow$: convergence in distribution) for the **whole path**, whatever the law of the steps Billingsley (1999)
- Right panel: $P(\max_{t \le 1} W_n(t) > 1)$ rises from 0.220 ($n = 5$) to 0.316 ($n = 2000$), towards the reflection-principle value $2(1 - \Phi(1)) = 0.3173$ (Φ: standard Normal distribution function)
- A coarse grid misses the maximum between observations and convergence is slow, of order $1/\sqrt{n}$: daily closes understate the intraday range

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Brownian Motion: Definition and Properties

- **Standard Brownian motion**: $W_0 = 0$; independent increments; $W_t - W_s \sim N(0, t-s)$ for $s < t$; continuous paths
  - Existence: Wiener (1923); simulation on a grid: $W_{t+\Delta t} = W_t + \sqrt{\Delta t}\, Z$, $Z \sim N(0,1)$
- $\mathrm{Cov}(W_s, W_t) = \min(s,t)$; a Gaussian process and a **martingale**: $\mathbb{E}[W_t \mid \mathcal{F}_s] = W_s$, with $\mathcal{F}_s$ the information up to s (the efficient-market random walk of Chapter 2)
- **Self-similarity**: $\{c^{-1/2} W_{ct}\}$ is again a standard Brownian motion for any $c > 0$
- Paths are continuous but **nowhere differentiable**
  - $(W_{t+h} - W_t)/h$ has standard deviation $1/\sqrt{h} \to \infty$: ordinary calculus fails

Brownian Paths (1/2)



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Brownian Paths (2/2)

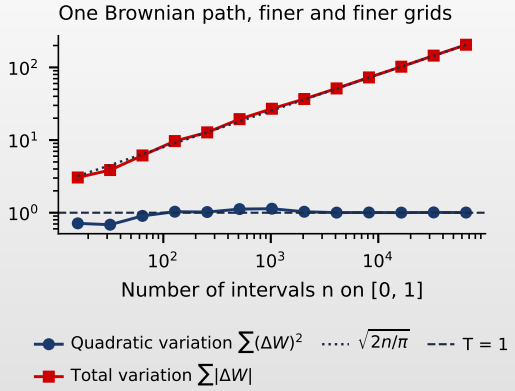
- Left: five paths with the pointwise band $\pm 1.96\sqrt{t}$ (95% for each t separately, not for the whole path); right: a path and its first 1% of time, stretched 100 times in time and 10 times in space
- The band widens like $\sqrt{t}$ (the square-root-of-time rule of Chapter 7); a zoomed piece looks like the whole path
  - Real markets break self-similarity: fat tails and clustering depend on the horizon (Chapter 1)

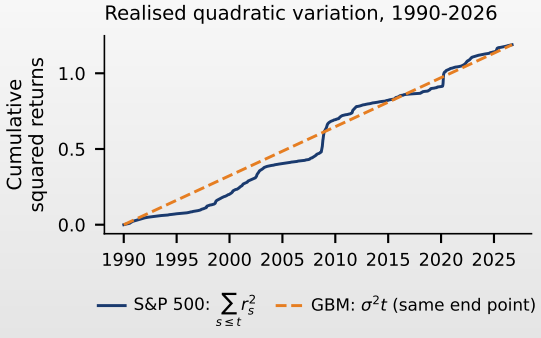
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Total and Quadratic Variation

- Grid $0 = t_0 < t_1 < \cdots < t_n = T$ with step $\Delta t = T/n$ and increments $\Delta W_k = W_{t_k} - W_{t_{k-1}}$
- **Total variation** $\sum_k |\Delta W_k|$: the length of the path
  - Its mean is $\sqrt{2nT/\pi} \to \infty$: a Brownian path has infinite length
- **Quadratic variation** $[W]_T = \lim \sum_k (\Delta W_k)^2 = T$
  - A random sum with a **deterministic** limit
  - $Q_n = \sum_{k=1}^{n} (\Delta W_k)^2$, $\mathrm{Var}(Q_n) = 2T^2/n$: one year of daily data gives about 9% relative error ($\sqrt{2/252}$), the idea behind realised variance (Chapter 9)
- Shorthand used in Itô calculus: $(\mathrm{d}W_t)^2 = \mathrm{d}t$

Quadratic Variation: Simulation and Data (1/2)





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Quadratic Variation: Simulation and Data (2/2)

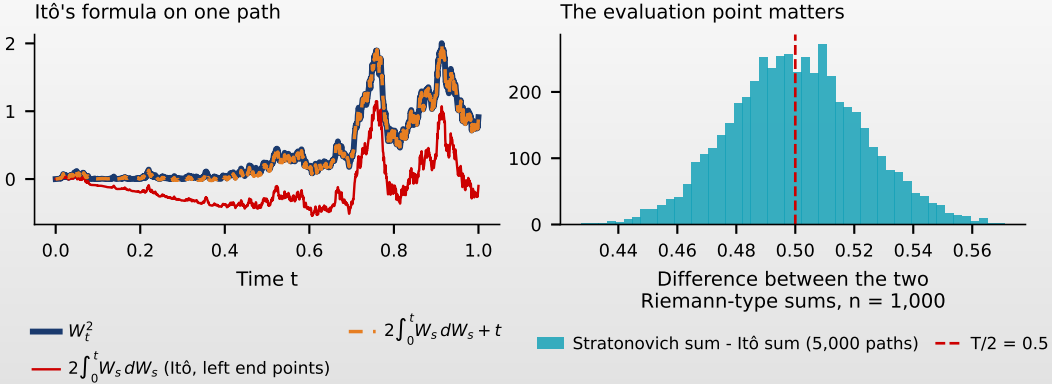
- Left: one Brownian path on grids with 16 to 65,536 intervals: the quadratic variation settles at 1.006 ($T = 1$), while the total variation keeps growing, 205 against $\sqrt{2n/\pi} = 204$
- Right: cumulative $\sum r_t^2$ of the S&P 500, 1.188 over 36.7 years (average volatility 18.0%); under geometric Brownian motion it would follow the straight line $\sigma^2 t$
- It rises in steps: September 2008 – March 2009 is 1.6% of the days but 15.2% of the quadratic variation; February – April 2020: 0.5% and 7.8%. Volatility is itself random (Heston, Section 10)

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The Itô Integral

- Goal: give a meaning to $\int_0^T H_t \, \mathrm{d}W_t$, the gain of holding H_t units of an asset whose price moves like W
  - Riemann–Stieltjes integration fails because the path has infinite total variation
- **Itô integral**: the limit in mean square of sums with the integrand at the **left** end point
  - $\int_0^T H_t \, \mathrm{d}W_t = \lim_{n\to\infty} \sum_{k=1}^{n} H_{t_{k-1}} \left(W_{t_k} - W_{t_{k-1}}\right)$, for H **adapted** with $\mathbb{E}\left[\int_0^T H_t^2 \, \mathrm{d}t\right] < \infty$
  - Financial reading: H_t uses only information up to t; the position is chosen before the next price move
- Zero mean, $\mathbb{E}\left[\int_0^T H_t \, \mathrm{d}W_t\right] = 0$, and the **Itô isometry**, $\mathbb{E}\left[\left(\int_0^T H_t \, \mathrm{d}W_t\right)^2\right] = \mathbb{E}\left[\int_0^T H_t^2 \, \mathrm{d}t\right]$
  - $t \mapsto \int_0^t H_s \, \mathrm{d}W_s$ is a martingale: no strategy earns money from a martingale on average (weaker conditions: ▸ Appendix)
- Sums with the integrand averaged over the two end points, $\frac{1}{2}(H_{t_{k-1}} + H_{t_k})$, give the **Stratonovich** integral, a different number; finance uses Itô, because that position would use the price at t_k, i.e. look into the future
  - For a smooth $f(W)$, evaluating at the mid-point $\frac{1}{2}(t_{k-1} + t_k)$ gives the same limit, but needs a grid twice as fine

Worked Example: $\int_0^T W_t \, \mathrm{d}W_t$ on Simulated Paths (1/2)



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Worked Example: $\int_0^T W_t \, \mathrm{d}W_t$ on Simulated Paths (2/2)

- Each simulated path uses 1 000 time steps.
- Each term: $W_{t_{k-1}} \Delta W_k = \frac{1}{2}(W_{t_k}^2 - W_{t_{k-1}}^2) - \frac{1}{2}(\Delta W_k)^2$; summed over k, the first parts give $\frac{1}{2}(W_T^2 - W_0^2) = \frac{1}{2} W_T^2$, because the intermediate terms cancel, and the second parts tend to $\frac{1}{2} T$: $\int_0^T W_t \, \mathrm{d}W_t = \frac{1}{2} W_T^2 - \frac{1}{2} T$
  - Ordinary calculus would give $\frac{1}{2} W_T^2$: the extra $-\frac{1}{2} T$ is the quadratic variation, and the mean is $\frac{1}{2}(T - T) = 0$
- Left: on one path, $2 \int_0^t W \, \mathrm{d}W + t$ lies exactly on W_t^2; right: Stratonovich minus Itô sums over 5,000 paths, mean 0.500, s.d. 0.022 (theory $T/2 = 0.5$)
- Itô sums: mean 0.007, variance 0.520; theory: 0 and, by the isometry, $\int_0^1 t \, \mathrm{d}t = 1/2$

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Itô Processes and Itô's Lemma

- **Itô process**: $\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t$ (drift a_t, diffusion coefficient b_t, both adapted)
  - Shorthand for $X_t = X_0 + \int_0^t a_s\,\mathrm{d}s + \int_0^t b_s\,\mathrm{d}W_s$
- **Itô's lemma**: for $f(t,x)$ twice differentiable Itô (1944)
  - $\mathrm{d}f(t,X_t) = \left(f_t + a_t f_x + \frac{1}{2}b_t^2 f_{xx}\right)\mathrm{d}t + b_t f_x\,\mathrm{d}W_t$
  - Partial derivatives evaluated at (t,X_t): $f_t = \partial f/\partial t$, $f_x = \partial f/\partial x$, $f_{xx} = \partial^2 f/\partial x^2$
- Rule of thumb: Taylor expansion to second order with $\mathrm{d}t\cdot\mathrm{d}t = 0$, $\mathrm{d}t\cdot\mathrm{d}W = 0$, $\mathrm{d}W\cdot\mathrm{d}W = \mathrm{d}t$
  - The second-order term $\frac{1}{2}b_t^2 f_{xx}\,\mathrm{d}t$ does not vanish: this is the whole novelty

Quadratic Covariation and the Multivariate Itô Formula

- **Quadratic covariation** $[X, Y]_t = \lim \sum_k \Delta X_k \, \Delta Y_k$; if $\mathrm{Corr}(\mathrm{d}W^{(1)}, \mathrm{d}W^{(2)}) = \rho$, then $\mathrm{d}[W^{(1)}, W^{(2)}]_t = \rho \, \mathrm{d}t$
  - Product rule: $\mathrm{d}(X_t Y_t) = X_t \, \mathrm{d}Y_t + Y_t \, \mathrm{d}X_t + \mathrm{d}[X, Y]_t$
- **Multivariate Itô formula** for $X_t \in \mathbb{R}^d$ and $f(t, x)$ twice differentiable
  - $\mathrm{d}f(t, X_t) = f_t \, \mathrm{d}t + \sum_i f_{x_i} \, \mathrm{d}X_t^i + \frac{1}{2} \sum_{i,j} f_{x_i x_j} \, \mathrm{d}[X^i, X^j]_t$
  - X_t^i: the i-th component; f_{x_i}, $f_{x_i x_j}$: first and second partial derivatives in those components
- Worked example (Heston, Section 10): $\mathrm{d}S = \mu S \, \mathrm{d}t + \sqrt{v} \, S \, \mathrm{d}W^{(1)}$, $\mathrm{d}v = \kappa(\theta - v) \, \mathrm{d}t + \xi \sqrt{v} \, \mathrm{d}W^{(2)}$
  - S: price, v: its variance; μ: drift, κ: speed of mean reversion, θ: long-run variance, ξ: volatility of the variance
  - $\mathrm{d}(S_t v_t) = S_t \, \mathrm{d}v_t + v_t \, \mathrm{d}S_t + \rho \, \xi \, v_t S_t \, \mathrm{d}t$
  - ρ enters the drift of the product: with $\rho < 0$, price and variance tend to move in opposite directions

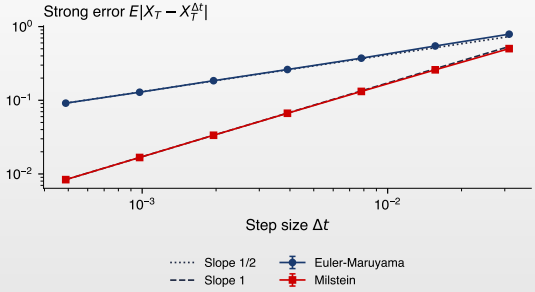
Stochastic Differential Equations and Two Schemes

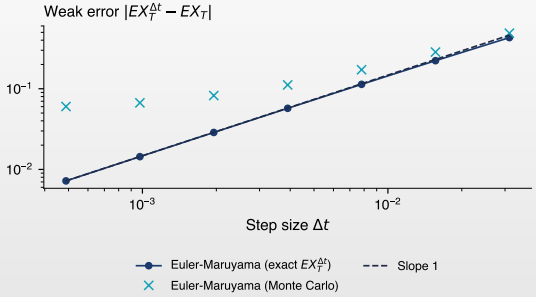
- A **stochastic differential equation** (SDE): $\mathrm{d}X_t = a(t, X_t)\,\mathrm{d}t + b(t, X_t)\,\mathrm{d}W_t$, X_0 given: drift and diffusion depend on the current state
  - Exact transitions for GBM, Ornstein–Uhlenbeck and Merton: simulate them exactly, with no discretisation error; Heston has only a closed-form characteristic function; existence conditions: Appendix
- **Euler–Maruyama**: freeze a and b over each step Higham (2001)
  - $X_{k+1} = X_k + a(X_k)\,\Delta t + b(X_k)\,\Delta W_k$, $\Delta W_k = \sqrt{\Delta t}\,Z_k$
  - X_k: the approximation at $t_k = k\Delta t$; Z_k: i.i.d. $N(0,1)$ draws
- **Milstein**: add the next Itô–Taylor term Milstein (1975)
  - $X_{k+1} = X_k + a(X_k)\,\Delta t + b(X_k)\,\Delta W_k + \frac{1}{2}\,b(X_k)\,b'(X_k)\left[(\Delta W_k)^2 - \Delta t\right]$
  - $b'(x) = \mathrm{d}b/\mathrm{d}x$; the correction has conditional mean zero: it raises the strong order (path accuracy), not the weak order

Strong and Weak Convergence

- **Strong error**: $\mathbb{E}|X_T - X_T^{\Delta t}| \leq C\,\Delta t^{\gamma}$, how close each simulated path is to the true one
  - X_T: exact solution at T; $X_T^{\Delta t}$: the scheme with step Δt; C: a constant; γ: the strong order
  - Euler–Maruyama $\gamma = 1/2$; Milstein $\gamma = 1$; needed for path-dependent quantities: hedging errors, barrier crossings
- **Weak error**: $|\mathbb{E}[g(X_T^{\Delta t})] - \mathbb{E}[g(X_T)]| \leq C\,\Delta t^{\beta}$, how close the distribution is
  - g: a function of the final value, e.g. an option pay-off; β: the weak order
  - Both schemes: $\beta = 1$; enough for prices of European options and other expected values
  - Conditions: smooth coefficients with bounded derivatives and a smooth g Kloeden & Platen (1992)
  - Square-root coefficients (CIR (Cox–Ingersoll–Ross), Heston), quantiles such as Value-at-Risk (VaR) and barrier crossings can converge more slowly
- Test equation Higham (2001): $\mathrm{d}X = \lambda X\,\mathrm{d}t + \mu X\,\mathrm{d}W$, $\lambda = 2$, $\mu = 1$, $X_0 = 1$, $T = 1$ (here λ, μ are only its two coefficients), with the exact solution as benchmark
  - On a log-log plot, the order is the slope of the error against the step size (next slide)

Convergence of the Schemes (1/2)





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Convergence of the Schemes (2/2)

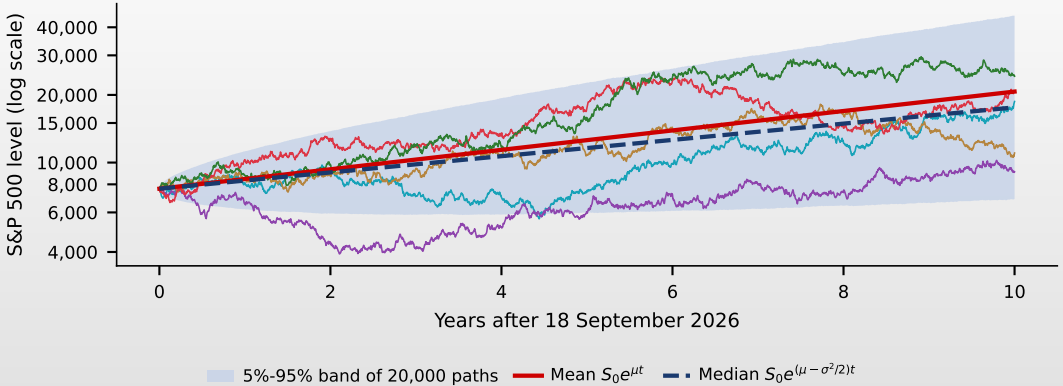
- 20,000 Brownian paths on a fine grid of 2^{11} steps, coarsened to steps from 2^{-11} to 2^{-5}
  - Strong orders (slope of the log-log fit, 95% CI by resampling whole paths): Euler–Maruyama 0.518 $[0.512, 0.525]$, Milstein 0.986 $[0.984, 0.987]$
  - The narrow CIs measure Monte Carlo noise only; the small gap to 1/2 and 1 is finite-step curvature, not a different order
  - Halving Δt halves the Milstein error, but cuts the Euler error only by 29%
- Weak order of Euler–Maruyama from the exact mean $(1 + \lambda \Delta t)^{T/\Delta t}$: 0.985, a deterministic slope
  - The Monte Carlo estimates flatten at small steps: their standard error, about 0.068, exceeds the bias
  - The sample size, not the step, is often the binding constraint Glasserman (2003)

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Geometric Brownian Motion (GBM) from Itô's Lemma

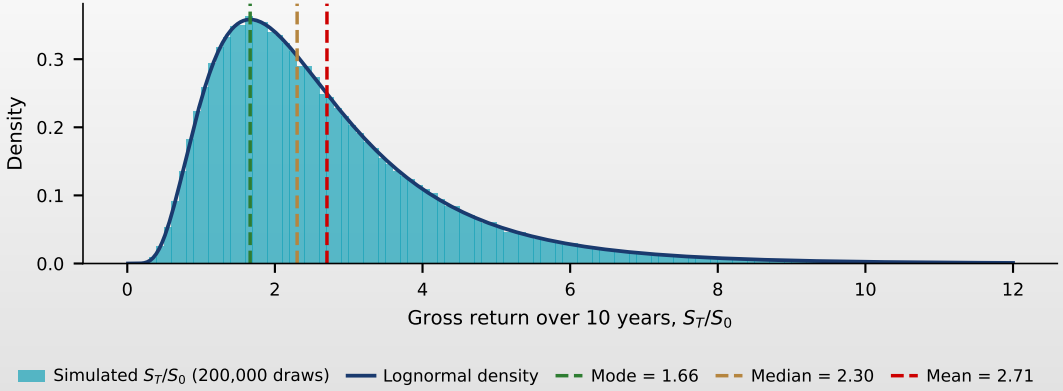
- SDE: $\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t$: the **percentage** change has constant drift μ and volatility σ per year
  - Proposed by Osborne (1959) and Samuelson (1965); the price model behind Black & Scholes (1973)
- Itô's lemma for $f(x) = \ln x$, with $f_x = 1/x$, $f_{xx} = -1/x^2$, $a_t = \mu S_t$, $b_t = \sigma S_t$
  - $\mathrm{d}\ln S_t = \left(\mu - \frac{1}{2}\sigma^2\right)\mathrm{d}t + \sigma\,\mathrm{d}W_t$, so $S_t = S_0 \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right)$
  - Prices stay positive; exact simulation needs only Normal draws
- The log price grows at the log drift $m = \mu - \sigma^2/2$, not at μ: volatility lowers the growth of the median investor's wealth
  - $\sigma = 18\%$ costs $\sigma^2/2 = 1.6$ percentage points a year; $\sigma = 67\%$ (Bitcoin), 22 points
- Testable implications: $r_t = \ln(S_t/S_{t-\Delta t})$ i.i.d. $N(m\Delta t, \sigma^2\Delta t)$ at every frequency
  - Skewness 0, excess kurtosis 0; no autocorrelation in r_t, $|r_t|$ or r_t^2; variance proportional to the horizon

Lognormal Outcomes: Ten Years of Simulated S&P 500 Paths (1/3)



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Lognormal Outcomes: Ten Years of Simulated S&P 500 Paths (2/3)



MFM_ch11_gbm_fan

Lognormal Outcomes: Ten Years of Simulated S&P 500 Paths (3/3)

- GBM with $\hat{\mu} = 10.0\%$, $\hat{\sigma} = 18.0\%$ (1990–2026), $T = 10$ years: mean $\mathbb{E}[S_T] = S_0 e^{\mu T}$, median $S_0 e^{mT}$, mode $S_0 e^{(m-\sigma^2)T}$; gross returns 2.71, 2.30, 1.66 (density plot)
- Fan chart: 20,000 exact paths from $S_0 = 7\,650$ (18 September 2026): mean 20 696, median 17 604, 5%–95% range 6 878 – 45 017; $P(S_T < S_0) = \Phi(-m\sqrt{T}/\sigma) = 7.1\%$
- $P(S_T < \mathbb{E}[S_T]) = \Phi(\sigma\sqrt{T}/2) = 61.2\%$: most investors end below the average, so report the median and quantiles; the fan is too narrow in crises (no fat tails, no clustering: tested below)
- The density uses 200 000 draws of S_T/S_0; the fan chart uses 20 000 paths.

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Estimating GBM by Maximum Likelihood

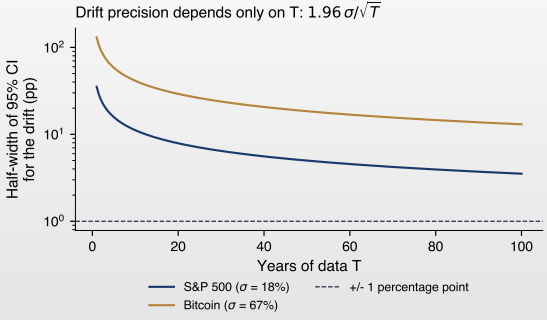
- Maximum likelihood estimation (MLE) from n log returns at step Δt, over $T = n\Delta t$ years ($\bar{r}$: mean log return): $\hat{\sigma}^2 = \frac{1}{n\Delta t}\sum_t (r_t - \bar{r})^2$, $\hat{m} = \bar{r}/\Delta t$, $\hat{\mu} = \hat{m} + \hat{\sigma}^2/2$
  - Under GBM, the standard error (s.e.) $\mathrm{s.e.}(\hat{\sigma}) = \sigma/\sqrt{2n}$ improves with more observations; $\mathrm{s.e.}(\hat{m}) = \sigma/\sqrt{T}$ depends only on the **calendar span**, since $\sum_t r_t = \ln(S_T/S_0)$ uses only the first and the last price

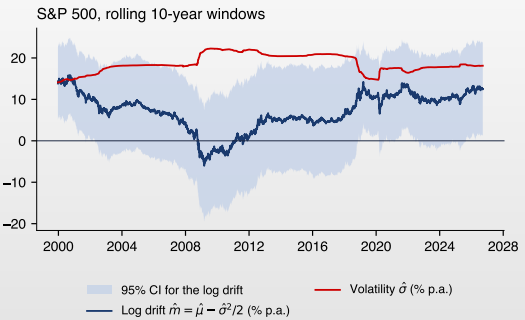
	Years	$\hat{\sigma}$ (%)	s.e.($\hat{\sigma}$)	$\hat{m}$ (%)	s.e.($\hat{m}$)
S&P 500	36.7	18.0	0.13	8.3	3.0
BET	26.5	22.2	0.19	16.1	4.3
Bitcoin	12.0	66.6	0.71	43.1	19.2

- Volatility is pinned down to 0.1–0.7 percentage points, the log drift only to 3–19 points; S&P 500: $\hat{\mu} = 8.3\% + 1.62\% = 10.0\%$ per year
- These s.e. are conditional on GBM: with excess kurtosis k, even independent returns inflate $\mathrm{s.e.}(\hat{\sigma})$ by $\sqrt{(k+2)/2} = 2.5$ (S&P 500); block-bootstrap s.e. in Seminar 11, B1

How Precisely Can We Estimate the Drift? (1/2)

p.p. = percentage points.





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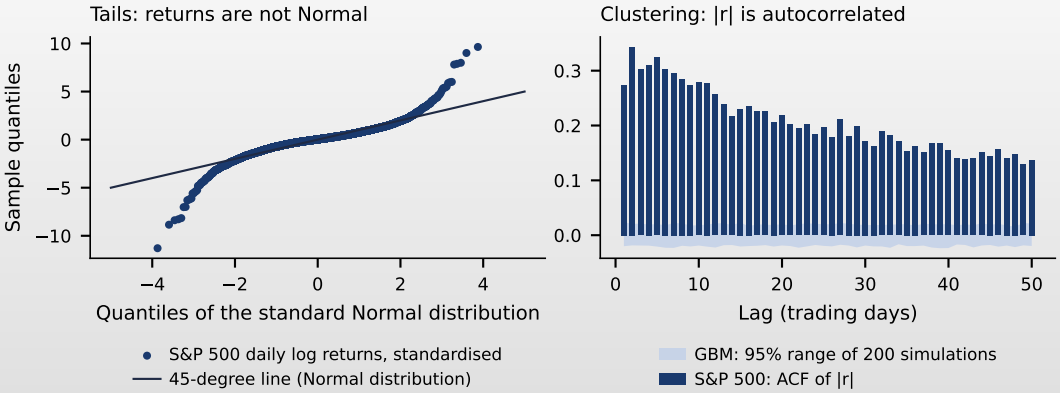
How Precisely Can We Estimate the Drift? (2/2)

- Left: half-width of the 95% CI for the drift against years of data; ± 1 point needs $(1.96\,\hat{\sigma}/0.01)^2 = 1\,243$ years for the S&P 500 and 17 028 years for Bitcoin
- Right: S&P 500 on rolling 10-year windows
  - The log drift ranges from -6.0% (window ending 2009-03) to 15.8% (2000-09), each with a CI of about ± 11 points, so the variation is mostly noise
  - Volatility stays between 14.1% and 22.3%: precise, but not constant
- Option pricing in Chapter 12 avoids μ: under the risk-neutral measure only σ enters (Girsanov, Section 8)

MFM_ch11_drift_precision

Does GBM Fit the S&P 500? (1/2)

ACF: AutoCorrelation Function.



MFM_ch11_gbm_vs_data

Does GBM Fit the S&P 500? (2/2)

- Left: quantile–quantile (QQ) plot of standardised daily log returns against the Normal distribution
  - Excess kurtosis 10.9 instead of 0; worst day at -11.3 standard deviations; 62 days beyond ± 4 against 0.59 expected; Jarque–Bera (JB) statistic 45 805
- Right: autocorrelation function (ACF) of $|r_t|$: 0.27 at lag 1 and still 0.14 at lag 50
  - The 95% band from 200 GBM samples of the same length stays below 0.024 in absolute value
- GBM fails on both stylised facts of Chapter 1 Cont (2001); the rest of the chapter adds mean reversion, jumps and random volatility

MFM_ch11_gbm_vs_data

Recap: Growth, Lognormality and Drift Precision

- Formulas to keep
  - $\ln(S_T/S_0) \sim N(mT, \sigma^2 T)$ with $m = \mu - \frac{1}{2}\sigma^2$: the median grows at m, the mean $\mathbb{E}[S_T]$ at μ
  - $\text{s.e.}(\hat{\sigma}) = \sigma/\sqrt{2n}$ shrinks with the number of returns; $\text{s.e.}(\hat{m}) = \sigma/\sqrt{T}$ only with the years covered
- From the data: S&P 500 $\hat{\sigma} = 18.0\%$ (s.e. 0.13), but $\hat{m} = 8.3\%$ (s.e. 3.0), in percentage points per year
- Check: with $\sigma = 20\%$, what is the s.e. of the log drift after 10 years of data, and after 40 years?
  - Answer: $0.20/\sqrt{10} = 6.3$ and $0.20/\sqrt{40} = 3.2$ percentage points; four times the span only halves the error, whatever the sampling frequency

The Ornstein–Uhlenbeck (OU) Process

- SDE: $\mathrm{d}x_t = \kappa(\theta - x_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, $\kappa > 0$ Uhlenbeck & Ornstein (1930): long-run mean θ, speed of mean reversion κ, volatility σ
  - Above θ the drift is negative, below θ positive: a spring pulling x back; used for interest rates Vasicek (1977), volatility, spreads, commodity prices
- Itô's lemma on $e^{\kappa t}x_t$: $x_t = \theta + (x_0 - \theta)e^{-\kappa t} + \sigma \int_0^t e^{-\kappa(t-s)}\,\mathrm{d}W_s$, a Normal variable with
  - $\mathbb{E}[x_t \mid x_0] = \theta + (x_0 - \theta)e^{-\kappa t}$, $\mathrm{Var}(x_t \mid x_0) = \frac{\sigma^2}{2\kappa}\left(1 - e^{-2\kappa t}\right)$
  - Stationary distribution ($t \to \infty$): $N\left(\theta, \sigma^2/(2\kappa)\right)$
- **Half-life** $\ln 2/\kappa$: the time for an expected deviation from θ to halve
  - $\kappa = 0.5$ per year: half-life 1.39 years; unlike Brownian motion, the variance stops growing

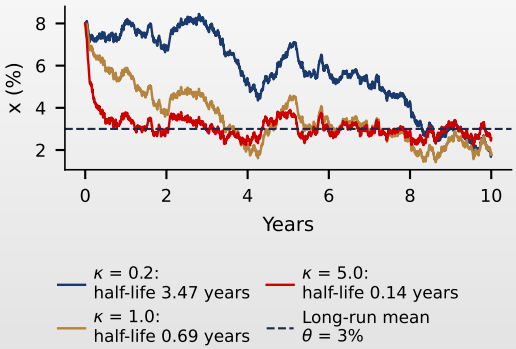
Solving the OU Equation Step by Step

- Step 1: apply Itô's lemma to $f(t,x) = e^{\kappa t}x$, with $f_t = \kappa e^{\kappa t}x$, $f_x = e^{\kappa t}$, $f_{xx} = 0$, so no second-order term appears
  - $\mathrm{d}(e^{\kappa t}x_t) = \kappa e^{\kappa t}x_t\,\mathrm{d}t + e^{\kappa t}\left[\kappa(\theta - x_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t\right] = \kappa\theta e^{\kappa t}\,\mathrm{d}t + \sigma e^{\kappa t}\,\mathrm{d}W_t$
  - The terms in x_t cancel: the right-hand side no longer depends on the unknown path
- Step 2: integrate from 0 to t, then multiply by $e^{-\kappa t}$
  - $e^{\kappa t}x_t - x_0 = \theta\left(e^{\kappa t} - 1\right) + \sigma\int_0^t e^{\kappa s}\,\mathrm{d}W_s$ gives the solution of the previous slide
- Step 3: the Itô integral has mean 0 and, by the isometry, variance $\sigma^2\int_0^t e^{-2\kappa(t-s)}\,\mathrm{d}s = \frac{\sigma^2}{2\kappa}\left(1 - e^{-2\kappa t}\right)$
- Check: $x_0 = 8\%$, $\theta = 3\%$, $\kappa = 0.5$, $\sigma = 1\%$: what are the mean and the standard deviation of x_1?
  - Answer: $3\% + 5\% \cdot e^{-0.5} = 6.03\%$ and $1\% \cdot \sqrt{1 - e^{-1}} = 0.80\%$: after one year 61% of the initial gap remains

Exact Discretisation: an AR(1)

- Sampled at step Δt the OU process is exactly an autoregressive model of order 1 (AR(1)):
  - $x_{t+\Delta t} = a + b\,x_t + \varepsilon_t$, $b = e^{-\kappa \Delta t}$, $a = \theta(1-b)$, $\mathrm{Var}(\varepsilon_t) = \frac{\sigma^2}{2\kappa}(1-b^2)$
- Maximum likelihood = ordinary least squares (OLS) of $x_{t+\Delta t}$ on x_t, then invert the mapping
  - $\hat{\kappa} = -\ln \hat{b}/\Delta t$, $\hat{\theta} = \hat{a}/(1-\hat{b})$, $\hat{\sigma} = \hat{s}_\varepsilon \sqrt{2\hat{\kappa}/(1-\hat{b}^2)}$
  - $\hat{a}$, $\hat{b}$: OLS intercept and slope; $\hat{s}_\varepsilon$: standard deviation of the OLS residuals
- Standard errors for $\hat{\kappa}$ and $\hat{\theta}$ by the delta method from the OLS covariance of $(\hat{a}, \hat{b})$
  - $\hat{\kappa}$ depends on $\hat{b}$ only: $\partial\kappa/\partial b = -1/(b\Delta t)$, so $\mathrm{Var}(\hat{b})$ suffices (worked case: Seminar 11, A1)
  - $\hat{\theta}$ has gradient $\big(1/(1-b),\ a/(1-b)^2\big)$ in (a, b), so it also needs $\mathrm{Var}(\hat{a})$ and $\mathrm{Cov}(\hat{a}, \hat{b})$
- With daily data b is very close to 1: κ is hard to estimate, as with unit-root tests

Three Speeds of Mean Reversion



- OU paths driven by the same shocks, starting at 8%
  - $\theta = 3\%$, $\sigma = 1\%$ and $\kappa = 0.2, 1, 5$; exact simulation, daily steps
- Half-lives 3.47, 0.69 and 0.14 years
  - With $\kappa = 5$ the gap from 8% to 3% closes within a few months
- Stationary standard deviations $\sigma/\sqrt{2\kappa}$: 1.58%, 0.71%, 0.32%
  - Slow reversion looks like a random walk for years
- Long-horizon forecasts converge to θ

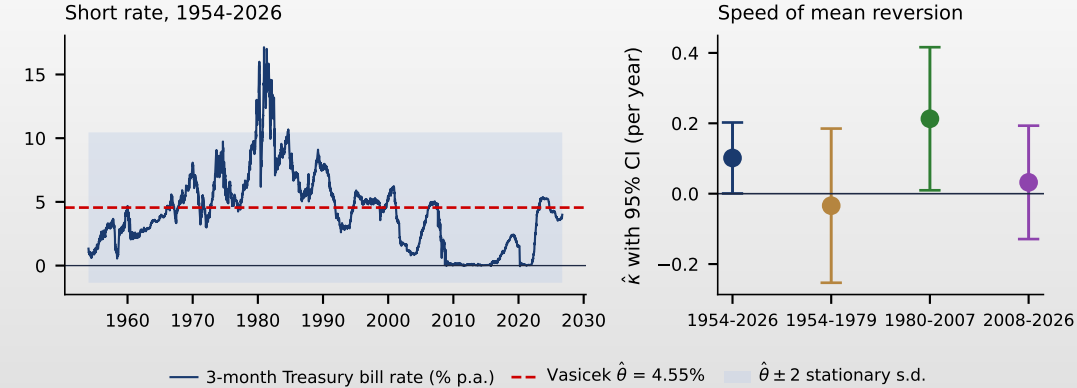
MFM_ch11_ou_paths

Short-Rate Models: Vasicek and CIR

- **Vasicek**: the short rate r_t is an OU process Vasicek (1977)
  - Normal distribution: negative rates have positive probability $\Phi(-\theta\sqrt{2\kappa}/\sigma)$ in the long run
  - Bond prices and yields in closed form (Feynman–Kac, Section 8)
- **Cox–Ingersoll–Ross** (CIR): $\mathrm{d}r_t = \kappa(\theta - r_t)\,\mathrm{d}t + \sigma\sqrt{r_t}\,\mathrm{d}W_t$ Cox, Ingersoll & Ross (1985)
  - Volatility shrinks near zero; the rate stays positive if $2\kappa\theta \geq \sigma^2$ (Feller condition) Feller (1951)
- Which is closer to the data is an empirical question, tested in Section 7 Chan, Karolyi, Longstaff & Sanders (1992)

Vasicek on the 3-Month Treasury Bill Rate (1/2)

s.d. = standard deviation.



MFM_ch11_vasicek

Vasicek on the 3-Month Treasury Bill Rate (2/2)

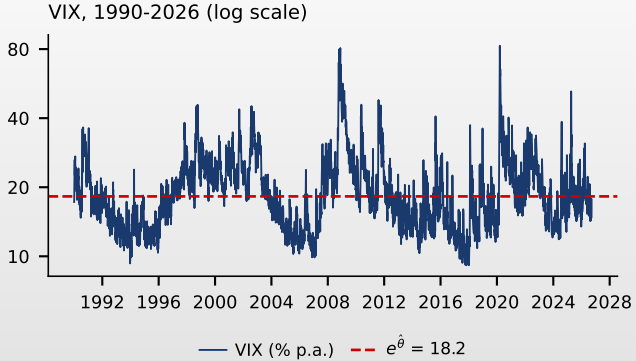
- Left: daily 3-month US Treasury bill rate (DTB3), 1954–2026, with the Vasicek long-run mean $\hat{\theta} = 4.55\%$
  - $\hat{\kappa} = 0.101$ (s.e. 0.051), half-life 6.8 years, $\hat{\sigma} = 1.33\%$, daily $\hat{b} = 0.99960$: almost a unit root
  - Band: $\hat{\theta}$ plus/minus two stationary standard deviations (s.d.)
- Right: $\hat{\kappa}$ with 95% CI by period: -0.034 (1954–1979, explosive), 0.213 (1980–2007), 0.032 (2008–2026); two of the three CIs include zero
- Under the simulated random-walk null, an estimate at least as large has probability 0.287: the sample does not establish mean reversion (Seminar 11, B2)

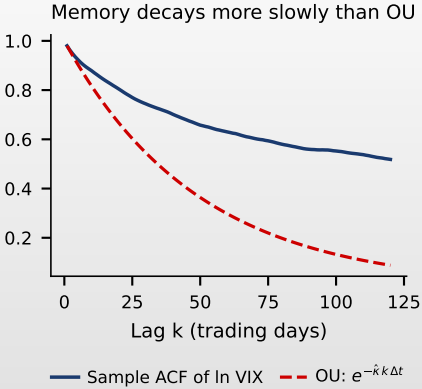
MFM_ch11_vasicek

Why Estimated Mean Reversion Looks Too Fast

- In finite samples $\hat{\kappa}$ overstates κ by roughly $4/T$ (T in years) Tang & Chen (2009)
  - Full-sample estimates as the truth, 500 samples of 72 years: mean $\hat{\kappa} = 0.164$, bias 0.063 against $4/T = 0.055$; 78% of the estimates above the truth
  - Estimated half-lives are too short and mean reversion looks faster than it is (Seminar 11, B2)
- Where the bias comes from: $\hat{b}$, the OLS slope of $x_{t+\Delta t}$ on x_t, is biased towards zero in a short persistent sample, and $\hat{\kappa} = -\ln \hat{b}/\Delta t$ turns a too-small $\hat{b}$ into a too-large $\hat{\kappa}$
  - The bias depends on the span T, not on the number of observations: sampling daily instead of monthly does not remove it (Section 7)

Is Volatility Mean-Reverting? The VIX (1/2)





MFM_ch11_vix_ou

Is Volatility Mean-Reverting? The VIX (2/2)

- Left: VIX 1990–2026 on a log scale with the OU long-run level $e^{\hat{\theta}} = 18.2$
  - OU fit to $\ln \mathrm{VIX}$: $\hat{\kappa} = 5.10$ (s.e. 0.53), half-life 34 trading days: spikes such as 82.7 (2020-03-16) fade within weeks
- Right: sample ACF of $\ln \mathrm{VIX}$ against the OU-implied $e^{-\hat{\kappa} k \Delta t}$: at lag 60, 0.63 in the data against 0.30 under OU: a fast decay, then a slow component
- One OU factor is too simple: candidates are two-factor or long-memory models, while rough volatility (Section 12) is a distinct alternative; mean-reverting volatility is the building block of Heston

MFM_ch11_vix_ou

Recap: Mean Reversion in Three Formulas

- Formulas to keep
  - Expected path: $\mathbb{E}[x_t \mid x_0] = \theta + (x_0 - \theta)e^{-\kappa t}$; half-life $\ln 2/\kappa$
  - Long-run spread: stationary standard deviation $\sigma/\sqrt{2\kappa}$
  - Estimation: AR(1) slope $b = e^{-\kappa \Delta t}$, hence $\hat{\kappa} = -\ln \hat{b}/\Delta t$, biased upwards in short samples
- From the data: DTB3 $\hat{\kappa} = 0.101$ (s.e. 0.051), a half-life of 6.8 years; for the log VIX the half-life is only 34 trading days
- Check: monthly data give $\hat{b} = 0.95$. What are $\hat{\kappa}$ and the half-life?
  - Answer: $\hat{\kappa} = -12 \ln 0.95 = 0.616$ per year and $\ln 2/0.616 = 1.13$ years, about 13.5 months

Infill versus Long-Span Asymptotics

Infill: $\Delta \to 0$ on a fixed $[0, T]$

- $\sum_k (\Delta X_k)^2 \to \int_0^T b_t^2 \, \mathrm{d}t$
- Diffusion coefficient: identified without error in the limit
  - Quadratic variation is observable (continuous Itô process, no measurement noise)
- Sampling more often helps σ (Section 5) and $b(x)$ (nonparametric, below)

Long span: $T \to \infty$

- The drift is not consistently estimable on a fixed span: its Fisher information grows with T, not with n
  - Fisher information $I(m) = -\mathbb{E}[\partial^2 \ell / \partial m^2]$ (ℓ: log-likelihood) bounds $\mathrm{Var}(\hat{m}) \geq 1/I(m)$; for n GBM returns $I(m) = n\Delta t/\sigma^2 = T/\sigma^2$
- GBM: $\mathrm{s.e.}(\hat{m}) = \sigma/\sqrt{T}$
- OU: $\mathrm{Var}(\hat{\kappa}) \approx (e^{2\kappa\Delta} - 1)/(T\Delta) \to 2\kappa/T$ as $\Delta \to 0$ (Seminar 11, A1)
- The bias of $\hat{\kappa}$ does not shrink with finer sampling Yu (2012)
  - $b = e^{-\kappa\Delta} \to 1$: $\hat{\kappa}$ is far from Normal when κT is small

- The imprecise GBM drift and the biased $\hat{\kappa}$ have the same cause: information grows with T, not with n

Likelihood for Nonlinear Diffusions (deeper)

- **Euler pseudo-likelihood**: a Gaussian transition $r_{t+\Delta} \mid r_t \sim N\left(r_t + a(r_t)\Delta,\ b^2(r_t)\Delta\right)$, with $a(r)$, $b(r)$ the drift and diffusion functions of the rate r and Δ the sampling step
  - Biased at order Δ: small with daily data, larger with monthly or quarterly steps
- **Exact CIR**: $2c\, r_{t+\Delta} \mid r_t$ is noncentral χ^2 with $4\kappa\theta/\sigma^2$ degrees of freedom Cox, Ingersoll & Ross (1985)
  - $c = 2\kappa/\left(\sigma^2(1 - e^{-\kappa\Delta})\right)$, noncentrality $2c\, r_t e^{-\kappa\Delta}$; Feller condition $\Leftrightarrow$ degrees of freedom ≥ 2 (Seminar 11, A7)
- General diffusions: closed-form Hermite expansions of the transition density Aït-Sahalia (2002)
- CKLS (Chan–Karolyi–Longstaff–Sanders): $\mathrm{d}r = \kappa(\theta - r)\,\mathrm{d}t + \sigma r^{\gamma}\,\mathrm{d}W$ nests Vasicek ($\gamma = 0$) and CIR ($\gamma = 1/2$) Chan, Karolyi, Longstaff & Sanders (1992)
  - Gaussian (Euler) quasi-likelihood; tests on γ with sandwich standard errors whose score covariance is heteroskedasticity- and autocorrelation-consistent (HAC) Newey & West (1987)

Vasicek, CIR and CKLS on the T-Bill Rate, 1954–2007 (deeper)

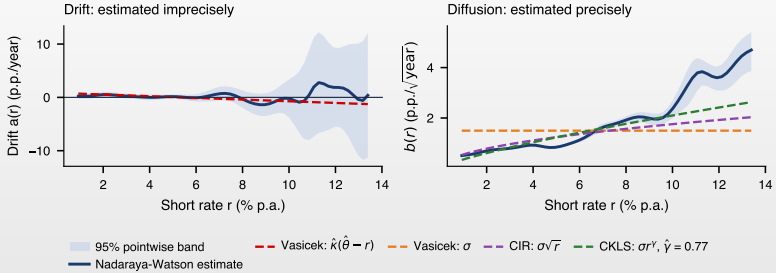
Model	$\hat{\kappa}$ daily (s.e.)	$\hat{\kappa}$ monthly (s.e.)	γ
Vasicek, exact	0.158 (0.073)	0.201 (0.083)	0
Vasicek, Euler	0.158 (0.073)	0.199 (0.082)	0
CIR, exact	0.113 (0.060)	0.153 (0.070)	1/2
CIR, Euler	0.093 (0.060)	0.144 (0.069)	1/2
CKLS, Euler	0.080 (0.057)	0.144 (0.068)	0.77 (0.033) daily; 0.60 (0.149) monthly

- Daily DTB3 ($n = 13\,487$) and month-end values ($n = 647$); 2008–2026 excluded: exact zeros and negative quotes lie outside the domain of these likelihoods (Appendix)
- Exact and Euler differ by less than one standard error: discretisation error is small next to the long-span uncertainty of $\hat{\kappa}$
- Volatility rises with the level: HAC Wald tests reject $\gamma = 0$ at both frequencies and $\gamma = 1/2$ with daily but not with monthly data (statistics: Appendix)

Nonparametric Drift and Diffusion (deeper)

- No functional form: kernel averages of the increments around each level x
  - $\Delta r_t = r_{t+\Delta} - r_t$: the next increment; $w_t(x)$: weight of day t, large when r_t is close to x
  - $\hat{a}(x) = \sum_t w_t(x)\,\Delta r_t/\Delta$, $\hat{b}^2(x) = \sum_t w_t(x)\,(\Delta r_t)^2/\Delta$, $w_t(x) = K\left((r_t - x)/h\right) / \sum_s K\left((r_s - x)/h\right)$
  - Nadaraya–Watson weights summing to one, Gaussian kernel K, bandwidth h Stanton (1997)
- $\mathbb{E}[\Delta r \mid r = x]/\Delta = a(x) + O(\Delta)$ and $\mathbb{E}[(\Delta r)^2 \mid r = x]/\Delta = b^2(x) + O(\Delta)$: exact only as $\Delta \to 0$
  - $O(\Delta)$: an error term of order Δ; the diffusion converges at the infill rate, the drift only at the long-span rate; valid for non-stationary but recurrent processes Bandi & Phillips (2003)
- Caution: nonparametric specification tests reject far too often with persistent data Pritsker (1998)

Nonparametric Drift and Diffusion of the T-Bill Rate



- Daily DTB3, 1954–2007; bandwidth $h = 0.44$ percentage points; 95% heteroskedasticity-robust pointwise bands (serial dependence ignored)
- Drift: zero lies inside the band at 97% of the levels; diffusion: median relative s.e. 8%, $b(r)$ rises from 0.5% to 4.7% (log-slope 0.90)
- No parametric shape fits everywhere: inside the band at 35% (CKLS), 7% (CIR), 5% (Vasicek) of the levels, a descriptive share

MFM_ch11_np_diffusion

Recap: What Frequency Buys and What It Cannot

- Formulas to keep
  - Infill: $\sum_k (\Delta X_k)^2 \to \int_0^T b_t^2 \, \mathrm{d}t$, so the diffusion coefficient is learned from frequent sampling
  - Long span: $I(m) = T/\sigma^2$ and $\mathrm{Var}(\hat{\kappa}) \approx 2\kappa/T$, so drift and speed are learned only from more years
  - Bias: $\hat{\kappa}$ exceeds κ by roughly $4/T$
- From the data: in simulated samples of 72 years the mean $\hat{\kappa}$ is 0.164, a bias of 0.063
- Check: the same 20 years are sampled daily instead of monthly (21 times more observations). Which standard error shrinks, that of $\hat{\sigma}$ or that of $\hat{\kappa}$?
  - Answer: that of $\hat{\sigma}$, by about $\sqrt{21} \approx 4.6$ times; that of $\hat{\kappa}$ barely moves, since $2\kappa/T$ does not depend on the step

Why Prices Use a Different Probability: a One-Period Example

- A stock costs 100 today and tomorrow is worth 120 (up) or 90 (down); the interest rate is 0; the real-world probability of up is $p = 0.6$
- A claim pays 1 if up and 0 if down. Replicate it with Δ shares and B in cash
  - $120\Delta + B = 1$ and $90\Delta + B = 0$ give $\Delta = 1/30$, $B = -3$: the replica costs $100/30 - 3 = 1/3$
  - Any other price is an arbitrage: sell the dearer of claim and replica, buy the cheaper, and keep the difference in both states
- The same price is an expectation under $q = (100 - 90)/(120 - 90) = 1/3$, the only weight with $120q + 90(1 - q) = 100$
  - Under q the stock is a martingale; p never entered: the expected payoff under P is 0.6, the price is $1/3$
- In general, absence of arbitrage means that discounted traded prices are martingales under an equivalent measure Q (fundamental theorem of asset pricing, Shreve (2004), Ch. 5); Girsanov builds this Q in continuous time
- Check: what is the price of the claim if $p = 0.9$? And if the down state is 80 instead of 90?
  - Answer: still $1/3$; with 80, $q = 20/40 = 1/2$, so the price is $1/2$

Girsanov's Theorem

- Real-world measure P and an equivalent measure Q (the same events of probability zero), linked by a **Radon–Nikodym density**
  - $\left.\frac{\mathrm{d}Q}{\mathrm{d}P}\right|_{\mathcal{F}_T} = Z_T = \exp\left(-\int_0^T \theta_s \,\mathrm{d}W_s - \frac{1}{2}\int_0^T \theta_s^2 \,\mathrm{d}s\right)$
  - $Z_T > 0$ with $\mathbb{E}^P[Z_T] = 1$ reweights the probability of each path: $Q(A) = \mathbb{E}^P[Z_T 1_A]$; θ_s: an adapted process (below, the market price of risk)
- Girsanov (1960): under Q, $\tilde{W}_t = W_t + \int_0^t \theta_s \,\mathrm{d}s$ is a Brownian motion if $\mathbb{E}\left[e^{\frac{1}{2}\int_0^T \theta_s^2 \mathrm{d}s}\right] < \infty$ (Novikov condition)
- For $\mathrm{d}X = a\,\mathrm{d}t + b\,\mathrm{d}W$: $\mathrm{d}X = (a - b\theta)\,\mathrm{d}t + b\,\mathrm{d}\tilde{W}$
  - For continuous Itô processes, **only the drift changes**: the diffusion coefficient and the quadratic variation are the same under P and Q
  - With jumps (Section 9), Q may also change the jump intensity and the jump-size distribution, a jump risk premium: $\hat{\lambda}$ from returns is not the intensity that prices options
  - Under GBM (constant σ), realised variance (Chapter 9) measures the σ that prices options, while μ drops out
  - With stochastic volatility, P and Q weigh future variance paths differently: historical variance is a benchmark, not a calibration

Market Price of Risk: P versus Q

- GBM: $\theta = (\mu - r)/\sigma$ is the **market price of risk**; under Q, $\mathrm{d}S = rS\,\mathrm{d}t + \sigma S\,\mathrm{d}\tilde{W}$ and $e^{-rt}S_t$ is a Q-martingale (Seminar 11, A4)
  - S&P 500, 1990–2026: $\hat{\theta} = (10.0\% - 2.7\%)/18.0\% = 0.40$, s.e. $1/\sqrt{T} = 0.17$ (r: mean 3-month T-bill rate): as imprecise as the drift
- Vasicek with risk premium $\lambda\sigma$: $\kappa^Q = \kappa$, $\kappa\theta^Q = \kappa\theta - \lambda\sigma$; here θ, θ^Q are long-run levels, not the market price of risk above
  - The long-run level that prices bonds differs from the one estimated on data (Section 6)
  - In this convention a positive term premium means $\lambda < 0$: then $\theta^Q = \theta - \lambda\sigma/\kappa > \theta$ and long yields are higher
- Heston with volatility risk premium λv_t (a separate λ): $\kappa^Q = \kappa + \lambda$, $\kappa^Q\theta^Q = \kappa\theta$
  - $\lambda < 0$ (investors pay for protection against variance) gives $\theta^Q > \theta$: the variance risk premium of Chapter 9

Feynman–Kac: From Expectations to the Pricing PDE

- If $\mathrm{d}X = a(X)\,\mathrm{d}t + b(X)\,\mathrm{d}\tilde{W}$ under Q Kac (1949)
  - $u(t,x) = \mathbb{E}^Q\left[e^{-\int_t^T r_s\,\mathrm{d}s}\, g(X_T) \mid X_t = x\right]$ solves $u_t + a\,u_x + \frac{1}{2}b^2 u_{xx} - r\,u = 0$, $u(T,x) = g(x)$
  - u: the price at t of a claim paying $g(X_T)$ at T; r_s: the short rate used to discount; u_t, u_x, u_{xx}: partial derivatives; PDE: partial differential equation
- Proof idea: $e^{-\int_0^t r_s\mathrm{d}s}\,u(t,X_t)$ is a Q-martingale, so Itô's lemma must give it a zero drift
- Vasicek bond ($g = 1$, $X = r$): $p(t,r;T) = \exp\left(A(\tau) - B(\tau)\,r\right)$, $\tau = T - t$ (Seminar 11, A6)
  - $B(\tau) = \frac{1-e^{-\kappa\tau}}{\kappa}$, $A(\tau) = \left(\theta^Q - \frac{\sigma^2}{2\kappa^2}\right)(B(\tau) - \tau) - \frac{\sigma^2 B(\tau)^2}{4\kappa}$
  - Yields $y(\tau) = (B(\tau)\,r - A(\tau))/\tau$ tend to $\theta^Q - \sigma^2/(2\kappa^2)$: rate uncertainty lowers long yields (convexity); comparison with Treasury yields: ▸ Appendix
- Chapter 12 derives Black–Scholes by the same route

Recap: Pricing under Q

- Formulas to keep
  - Girsanov: under Q the drift a becomes $a - b\theta$, the diffusion b is unchanged
  - GBM: $\theta = (\mu - r)/\sigma$ turns the drift μ into r, so $e^{-rt}S_t$ is a Q-martingale
  - Feynman–Kac: price $= \mathbb{E}^Q$[discounted pay-off], the solution of the pricing PDE
- From the data: S&P 500 market price of risk 0.40 per $\sqrt{\text{year}}$, s.e. 0.17
- Check: $\mu = 9\%$, $\sigma = 20\%$, $r = 3\%$. What are the market price of risk and the drift of S under Q?
  - Answer: $\theta = 0.06/0.20 = 0.30$ and $\mu - \sigma\theta = 9\% - 6\% = 3\% = r$, with $\sigma = 20\%$ unchanged

Merton's Jump-Diffusion

- Brownian paths are continuous, but prices gap on news: on 19 October 1987 the Dow Jones fell 22.6%, about 20 standard deviations under GBM with 18% volatility
  - A jump cannot be hedged by continuous trading Merton (1976)
- Log price = Brownian motion with drift + **compound Poisson** jumps: $\mathrm{d}\ln S_t = m\,\mathrm{d}t + \sigma\,\mathrm{d}W_t + \mathrm{d}J_t$, $J_t = \sum_{i \le N_t} Y_i$
  - N_t Poisson with intensity λ; sizes $Y_i \sim N(\mu_J, \sigma_J^2)$ i.i.d.; W, N and Y independent; five parameters $(m, \sigma, \lambda, \mu_J, \sigma_J)$
- Return over Δt given k jumps: $N\left(m\Delta t + k\mu_J,\ \sigma^2\Delta t + k\sigma_J^2\right)$; unconditionally a **Poisson mixture** of Normal densities, i.i.d. over time
  - $f(r) = \sum_{k\ge 0} e^{-\lambda\Delta t}\frac{(\lambda\Delta t)^k}{k!}\,\varphi\left(r;\ m\Delta t + k\mu_J,\ \sigma^2\Delta t + k\sigma_J^2\right)$
  - Weight: Poisson probability of k jumps; $\varphi(r; \text{mean}, \text{variance})$: the Normal density
- Exact simulation: draw $N \sim \mathrm{Poisson}(\lambda\Delta t)$, then draw from the Normal distribution with the conditional moments

Moments of the Merton Return

- Cumulants c_k of the return over Δt
  - Definition: $K(u) = \ln \mathbb{E}[e^{ur}]$ and $c_k = K^{(k)}(0)$: c_1 mean, c_2 variance, c_3 third central moment; cumulants of independent parts add, and a Normal part has $c_k = 0$ for $k \geq 3$
  - The k-th cumulant of a compound Poisson sum is $\lambda \Delta t \, \mathbb{E}[Y^k]$
  - $c_2 = \sigma^2 \Delta t + \lambda \Delta t(\mu_J^2 + \sigma_J^2)$, $c_3 = \lambda \Delta t(\mu_J^3 + 3\mu_J \sigma_J^2)$
  - $c_4 = \lambda \Delta t(\mu_J^4 + 6\mu_J^2 \sigma_J^2 + 3\sigma_J^4)$; skewness $c_3 / c_2^{3/2}$, excess kurtosis c_4 / c_2^2
- Negative μ_J gives negative skewness; any jump gives positive excess kurtosis
- Excess kurtosis falls like $1/\Delta t$ as the horizon grows: jumps matter most at short horizons
  - Example: $\sigma = 15\%$, $\lambda = 5$, $\mu_J = -3\%$, $\sigma_J = 4\%$: daily excess kurtosis 17.6, skewness -2.07 (Seminar 11, A3)

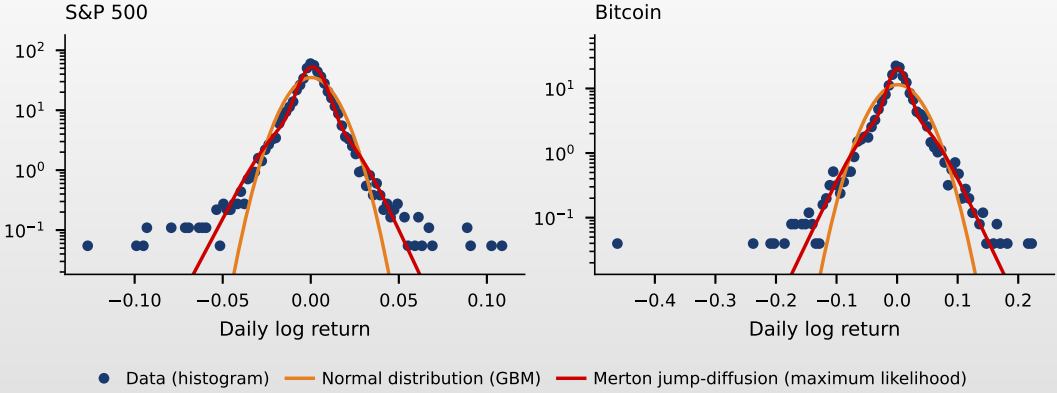
Merton Estimates for the S&P 500, BET and Bitcoin

LR (Likelihood Ratio): the likelihood-ratio statistic.

	$\hat{\sigma}$ (%)	$\hat{\lambda}$ (per year)	$\hat{\mu}_J$ (%)	$\hat{\sigma}_J$ (%)	Jump share of variance	LR
S&P 500	9.4	102 (11)	−0.16	1.44	71%	2 401
BET	11.3	79 (7)	−0.09	2.07	73%	2 326
Bitcoin	24.6	239 (21)	0.00	3.85	85%	1 357

- Log-likelihood maximised numerically over the Poisson mixture (up to 12 jumps per day); standard errors from the Hessian in brackets
- The likelihood is **unbounded**: with $m\Delta t$ equal to one observed return and $\sigma \to 0$ that density explodes, so the MLE is a local maximum Honoré (1998)
  - The LR test against GBM ($\lambda = 0$) is defined on $\sigma \geq 5\%$ and is not χ^2_3: $\lambda = 0$ lies on the boundary and μ_J, σ_J are unidentified; bootstrap critical values in Seminar 11, B5
- The fit prefers **many small** jumps (about 80–240 per year), not a few crashes

Merton Fit to the Return Distribution (1/2)



MFM_ch11_merton_density

Merton Fit to the Return Distribution (2/2)

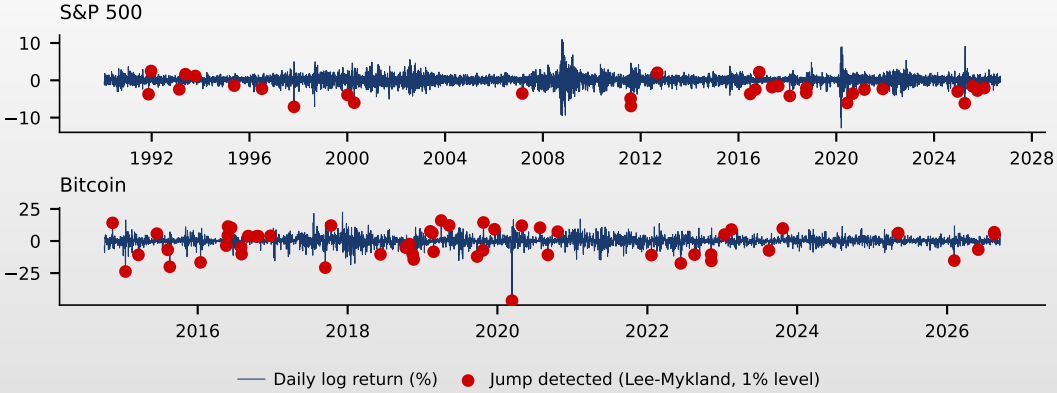
- Density of daily log returns on a log scale: histogram, the Normal distribution implied by GBM and the fitted Merton mixture
  - Merton follows the peak and the shoulders far better
- S&P 500: implied excess kurtosis 3.7 against 10.9 in the data, skewness -0.31 against -0.36; the most extreme days lie beyond both fitted curves
- About 102 "jumps" a year: the model uses jumps to mimic **changing volatility**; an i.i.d. model cannot tell a jump from a volatile period, so a direct test is needed

MFM_ch11_merton_density

Detecting Jumps: the Lee–Mykland Test

- Idea: a jump is a return that is huge **relative to the local volatility** Lee & Mykland (2008)
  - $L_t = r_t/\hat{\sigma}_t$, $\hat{\sigma}_t^2 = \frac{1}{K-2}\sum_{j=t-K+2}^{t-1} |r_j||r_{j-1}|$ (bipower variation, robust to jumps)
- Under no jumps, the maximum of $|L_t|$ over n days has a Gumbel limit
  - Reject if $(|L_t| - C_n)/S_n > -\ln(-\ln(1-\alpha))$; at $\alpha = 1\%$ this threshold for the **normalised** statistic, not for $|L_t|$, is 4.60
  - $C_n = \frac{\sqrt{2\ln n}}{c} - \frac{\ln\pi + \ln\ln n}{2c\sqrt{2\ln n}}$, $S_n = \frac{1}{c\sqrt{2\ln n}}$, $c = \sqrt{2/\pi}$ (the mean of $|Z|$, which rescales the bipower estimate), n = number of tested days
- Daily data: window $K = 16$ days, as recommended by the authors
- Controls, asymptotically, the probability of a false jump over the **whole** sample, not day by day

Jumps Detected in the S&P 500 and Bitcoin (1/2)



MFM_ch11_jumps

Jumps Detected in the S&P 500 and Bitcoin (2/2)

	S&P 500	Bitcoin
Candidates	32	52
Per year	0.87	4.35
Negative (%)	84	52
In the top 10 moves	0	5

- Daily log returns; days flagged at the 1% level
- S&P 500: the largest moves (2008, 2020) are not flagged
  - They came in turbulent periods; a 1.2% move in a calm market is flagged (1993-10-14)
- Detections differ from the fitted Merton intensity
  - Stochastic volatility; drift bursts (short explosive drift episodes)
  - Low detection power in daily data

MFM_ch11_jumps

Recap: What Jumps Add

- Formulas to keep
  - $\mathrm{d}\ln S_t = m\,\mathrm{d}t + \sigma\,\mathrm{d}W_t + \mathrm{d}J_t$: mean log return $m + \lambda\mu_J$ per year, so m alone is the drift of the diffusion part only
  - Higher cumulants come only from the jumps: $c_k = \lambda\Delta t\,\mathbb{E}[Y^k]$ for $k \geq 3$
  - Excess kurtosis c_4/c_2^2 falls like $1/\Delta t$ with the horizon
- From the data: S&P 500 $\hat{m} + \hat{\lambda}\hat{\mu}_J = 24.8\% + (-16.5\%) \approx 8.3\%$, close to the GBM log drift 8.3%; jump share of variance 71%
- Check: jumps give a daily excess kurtosis of 12. What is it for 10-day returns, if returns are i.i.d.?
  - Answer: $12/10 = 1.2$: over 10 days both c_4 and c_2 grow 10 times, so c_4/c_2^2 falls 10 times

The Heston Model

- Two SDEs, for the log price and for the instantaneous variance v_t Heston (1993)
  - $\mathrm{d}\ln S_t = \left(\mu - \frac{1}{2}v_t\right)\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^{(1)}$
  - $\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^{(2)}$, $\mathrm{Corr}(\mathrm{d}W^{(1)}, \mathrm{d}W^{(2)}) = \rho$
- Variance follows a CIR process: mean-reverting to θ with speed κ; ξ = volatility of volatility
  - Feller condition $2\kappa\theta \geq \xi^2$: variance never touches zero Feller (1951)
- $\rho < 0$: **leverage effect**, falling prices come with rising volatility
  - A fall raises the variance that scales the next moves, so returns are negatively skewed (Section 11)
  - Simulation: Euler scheme with full truncation $v^+ = \max(v, 0)$ Lord, Koekkoek & van Dijk (2010); an alternative discretisation, the quadratic-exponential scheme Andersen (2008)

Heston Parameters from the VIX

- Proxy $y_t = (\mathrm{VIX}_t/100)^2$ in place of v_t; discretised CIR regression without an extra intercept
  - $\frac{y_{t+\Delta t}-y_t}{\sqrt{y_t}} = \kappa\theta\frac{\Delta t}{\sqrt{y_t}} - \kappa\Delta t\sqrt{y_t} + \xi\sqrt{\Delta t}\,\varepsilon_t$; ρ = correlation of return and variance shocks
- S&P 500 and VIX, 1990–2026: $\hat{\kappa} = 4.13$ (half-life 42 days), $\hat{\xi} = 0.56$, $\hat{\rho} = -0.69$
  - $\sqrt{\hat{\theta}} = 20.9\%$ from the VIX, but the realised volatility is 18.0%; Feller ratio of the proxy $2\hat{\kappa}\hat{\theta}/\hat{\xi}^2 = 1.16$
- But y_t is **not** v_t: it is a risk-neutral expectation of future variance, affine in v_t (next slide)
  - For the simulated paths we set θ to the realised variance of the S&P 500; $\hat{\xi}$ is attenuated relative to ξ

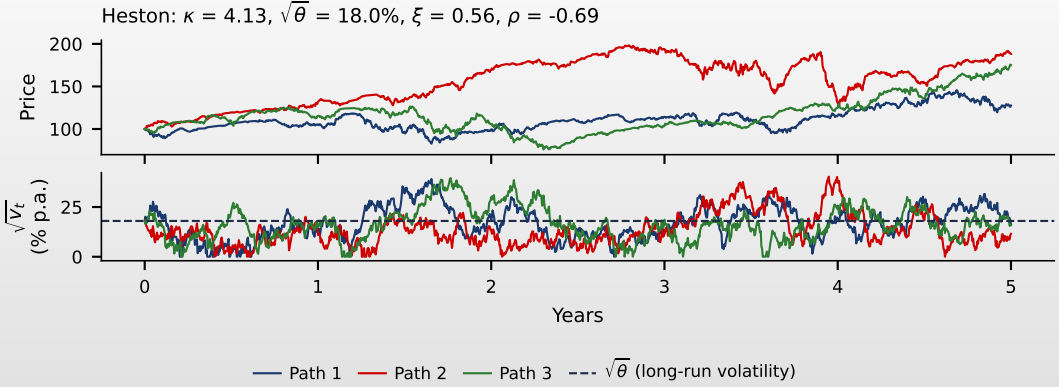
What the VIX Identifies in Heston (deeper)

- Under Q: $y_t = \frac{1}{\tau}\mathbb{E}^Q\left[\int_t^{t+\tau} v_s \, \mathrm{d}s \mid \mathcal{F}_t\right] = a + b\, v_t$, $\tau = 30/365$ (Seminar 11, A8)
  - $b = \frac{1-e^{-\kappa^Q \tau}}{\kappa^Q \tau} < 1$, $a = \theta^Q(1-b)$: **affine** in v_t, with Q-parameters
- What the regression on y_t recovers
  - κ: the P-speed, since an affine map keeps the dynamics of v_t
  - ρ: unchanged (correlation is scale-free), but it needs paired return and variance shocks, which daily proxy changes only approximate
  - ξ: attenuated, $\xi_y = b\,\xi$ at $v_t = \theta^Q$ (derivation: next slide; local correction $\hat{\xi}/b$: Appendix)
  - The long-run level $a + b\,\theta^P$ mixes θ^Q and θ^P: recover $\theta^P = (\theta_y - a)/b$, θ_y the estimated long-run level of y, before a Feller check on v_t
- Proper estimation fits the shifted square-root dynamics of y, or v_t and the observation equation jointly
- κ^Q is not identified from the VIX alone: joint P/Q estimation with option prices Chernov & Ghysels (2000), or moments of realised variance Bollerslev & Zhou (2002)

Where the Attenuation of ξ Comes From (deeper)

- Step 1: $y_t = a + b\,v_t$ with constants a, b, so $\mathrm{d}y_t = b\,\mathrm{d}v_t$ under P
  - $\mathrm{d}y_t = b\kappa(\theta^P - v_t)\,\mathrm{d}t + b\,\xi\sqrt{v_t}\,\mathrm{d}W_t^{(2)}$
- Step 2, drift: $b\kappa(\theta^P - v_t) = \kappa(\theta_y - y_t)$ with $\theta_y = a + b\,\theta^P$
  - The speed κ is unchanged; inverting gives $\theta^P = (\theta_y - a)/b$
- Step 3, diffusion: the CIR regression assumes $\xi_y\sqrt{y_t}$; matching $\xi_y\sqrt{y_t} = b\,\xi\sqrt{v_t}$ gives
  - $\xi_y = b\,\xi\sqrt{v_t/(a + b\,v_t)}$; at $v_t = \theta^Q$, $a + b\,\theta^Q = \theta^Q$, so $\xi_y = b\,\xi$
- Numbers: with $\kappa^Q = 5$, $b = 0.82$, so the fitted $\hat{\xi} = 0.56$ corresponds to $\hat{\xi}/b = 0.68$ near $v_t = \theta^Q$ (other κ^Q: Appendix)

Simulated Heston Paths (1/2)



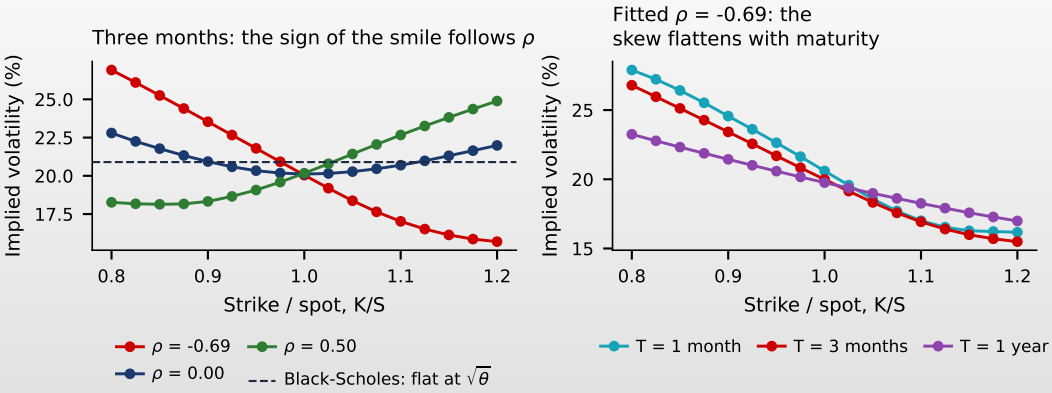
MFM_ch11_heston_paths

Simulated Heston Paths (2/2)

- Three five-year paths; top: price, bottom: $\sqrt{v_t}$
  - VIX-based $\hat{\kappa}$, $\hat{\xi}$ (uncorrected), $\hat{\rho}$; θ = realised S&P 500 variance
- Clustering now comes from the model
  - Price falls coincide with volatility spikes
  - Simulated return–variance correlation -0.69 (leverage through ρ)
- Feller ratio $2\hat{\kappa}\theta/\hat{\xi}^2 = 0.86 < 1$: the variance can reach zero
  - Negative values arise only in the Euler scheme
  - Full truncation sets them to zero (1.4% of the steps)

MFM_ch11_heston_paths

A Preview of Chapter 12: the Volatility Smile (1/2)



MFM_ch11_heston_smile

A Preview of Chapter 12: the Volatility Smile (2/2)

- **Implied volatility**: the σ that makes the Black–Scholes price equal to a given option price, flat across strikes under GBM; here from Heston call prices (Monte Carlo, 100,000 antithetic paths)
- Left, three months: with $\hat{\rho} = -0.69$, 26.9% at strike 80% of spot, 20.0% at the money, 15.7% at 120%: a downward **skew**; with $\rho = 0$ a symmetric **smile** (22.8%, 20.1%, 22.0%); with $\rho = 0.50$ the skew reverses, implied volatility rising with the strike from 18.3% to 24.9%
- Right: the skew flattens with maturity, from 27.9% – 16.2% at one month to 23.2% – 17.0% at one year; illustration only: pricing needs κ^{Q}, θ^{Q} calibrated to option prices (Chapter 12)

MFM_ch11_heston_smile

Affine Models and Conditional Volatility (deeper)

GARCH (Generalised Autoregressive Conditional Heteroskedasticity) models conditional variance in discrete data.

Affine jump-diffusions Duffie, Pan & Singleton (2000)

- Drift, variance, jump intensity affine in the state X_t
  - Vasicek, CIR, Merton, Heston and Bates
- $\mathbb{E}\left[e^{iu^\top X_T} \mid X_t\right] = \exp\left(\alpha(\tau) + \beta(\tau)^\top X_t\right)$
  - $i = \sqrt{-1}$, u: a real vector, $\tau = T - t$
  - α, β: solutions of Riccati ODEs
- Option prices by Fourier inversion
  - Simplest case: the Vasicek $A(\tau)$, $B(\tau)$ of Section 8

From GARCH to continuous time Nelson (1990)

- As the step shrinks, GARCH(1,1) converges to
  - $\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi v_t\,\mathrm{d}W_t^{(2)}$
  - Like Heston, with v_t instead of $\sqrt{v_t}$
- $\kappa \approx (1 - \alpha - \beta)/\Delta t$; α, β: GARCH(1,1) coefficients
  - Daily $\alpha + \beta = 0.99$: $\kappa \approx 2.5$ per year, half-life about 70 days (Chapter 5)
- GARCH and its diffusion limit are not asymptotically equivalent statistical experiments Wang (2002): inference can differ even as $\Delta t \to 0$

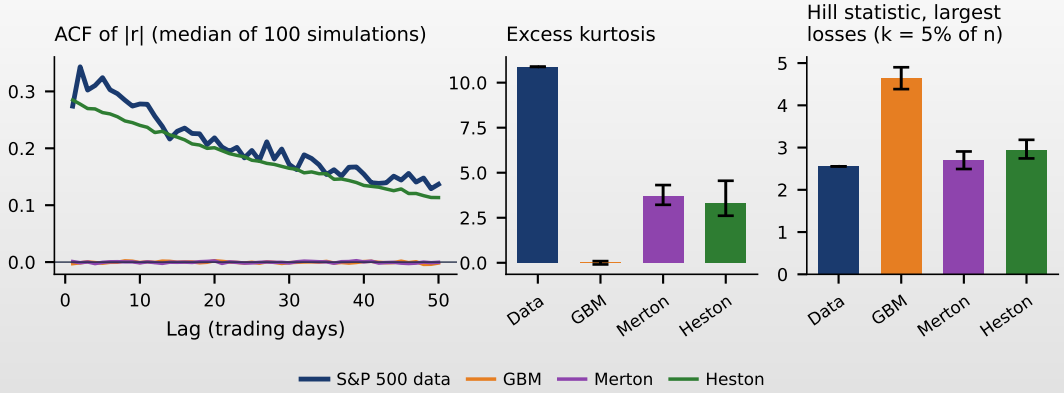
Recap: Stochastic Volatility

- Formulas to keep
  - Variance is a CIR process, $\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^{(2)}$, positive if $2\kappa\theta \geq \xi^2$
  - $\rho = \mathrm{Corr}(\mathrm{d}W^{(1)}, \mathrm{d}W^{(2)}) < 0$ gives leverage, negatively skewed returns and a downward implied-volatility skew
  - $(\mathrm{VIX}/100)^2 = a + b\,v_t$ with Q-parameters: κ and ρ carry over, ξ is attenuated by b
- From the data: $\hat{\rho} = -0.69$ and $\hat{\kappa} = 4.13$, a variance half-life of 42 trading days
- Check: with $\kappa = 4$ and $\theta = 0.04$, what is the largest ξ for which the Feller condition holds?
  - Answer: $\xi \leq \sqrt{2 \cdot 4 \cdot 0.04} = \sqrt{0.32} \approx 0.57$; a larger vol-of-vol lets the variance touch zero

Three Yardsticks for Simulated Returns

- **Shape**: skewness $c_3/c_2^{3/2}$ and excess kurtosis c_4/c_2^2 of daily returns (cumulants: Section 9)
- **Clustering**: the ACF of $|r_t|$ at lags 1–20; it is zero for any i.i.d. model, Merton included
- **Tail**: Hill statistic of the k largest losses $L_{(1)} \geq \cdots \geq L_{(k+1)}$, $\hat{\alpha} = \left[\frac{1}{k}\sum_{i=1}^{k} \ln\left(L_{(i)}/L_{(k+1)}\right)\right]^{-1}$
  - It estimates α in $P(L > x) \approx Cx^{-\alpha}$ for large x: $P(L > 2x)/P(L > x) \approx 2^{-\alpha}$
  - With $\alpha = 2.56$, doubling the loss size leaves $2^{-2.56} \approx 0.17$ of the frequency; a Normal tail leaves $P(Z > 4)/P(Z > 2) \approx 0.0014$: a lower α is a heavier tail
  - The estimate depends on k: for the data, 2.97, 2.56 and 2.04 at $k = 2.5\%$, 5% and 10% of the returns
- Check: losses twice as large are 8 times rarer. What is α?
  - Answer: $2^{-\alpha} = 1/8$, so $\alpha = 3$

Which Model Reproduces the S&P 500? (1/2)



MFM_ch11_models_vs_data

Which Model Reproduces the S&P 500? (2/2)

- 100 simulated samples of 9 245 days from each fitted model (medians, 90% ranges)
  - ACF of $|r_t|$, excess kurtosis, Hill statistic of the largest losses (k = 5% of the returns; lower = heavier tail)
  - Data: excess kurtosis 10.88, Hill 2.56
- Merton gets the Hill statistic (2.70) and skewness (-0.32 against -0.36) but no clustering; Heston, with the VIX-based parameters, gets the clustering
- The simulated Heston skewness is 0.00 by construction: the simulation uses a one-day Euler step, while continuous-time Heston implies about -0.20 at these parameters (next slide)
- No fitted model reaches the kurtosis of the data (full table: Appendix)
  - Andersen, Benzoni & Lund (2002) and Eraker, Johannes & Polson (2003) favour jumps **and** stochastic volatility (SVJ, stochastic volatility with jumps, Bates (1996))

MFM_ch11_models_vs_data

Why a Negative ρ Skews Daily Returns (deeper)

- Over one day $h = 1/252$, ignoring drift terms of order h: $R = \int_0^h \sqrt{v_s}\,\mathrm{d}W_s^{(1)}$
  - Itô's lemma on $\sqrt{v}$: $\mathrm{d}\sqrt{v_s} = \frac{1}{2}\xi\,\mathrm{d}W_s^{(2)} + O(\mathrm{d}s)$, so $\sqrt{v_s} \approx \sqrt{v_0} + \frac{1}{2}\xi W_s^{(2)}$
  - $R \approx \sqrt{v_0}\,W_h^{(1)} + \frac{1}{2}\xi \int_0^h W_s^{(2)}\,\mathrm{d}W_s^{(1)}$
- Third moment: the cross term gives $\mathbb{E}[R^3] \approx \frac{3}{2}\,\xi\rho\,v_0 h^2$, while $\mathrm{Var}(R) \approx v_0 h$
  - Skewness $\approx \frac{3}{2}\,\rho\xi\sqrt{h}/\sqrt{v_0}$: with $\hat{\rho} = -0.69$, $\hat{\xi} = 0.56$ and $\sqrt{v_0} = 18.0\%$, about -0.20, a leading-order approximation

Which Model When? Features, Identification and Limits

Feature in the data	Model	What the data identify	Unresolved
Positive prices, $\sqrt{t}$ scaling	GBM	σ precisely; μ only over very long spans	fat tails, clustering
Mean reversion of rates and of the VIX	OU, Vasicek, CIR, CKLS	$b(r)$ precisely; κ biased upwards, needs a long span	one factor too simple for the VIX
Gaps, short-horizon fat tails	Merton	λ, σ_J only under i.i.d. returns; the jump share depends on the frequency	no clustering
Clustering, leverage, implied-volatility skew	Heston	κ, ρ from the VIX; ξ attenuated; $\theta^Q \neq \theta^P$	kurtosis of the data
All of the above	Bates (Heston + jumps)	returns and option surfaces jointly	latent states: filtering needed

- Roughness and long memory of volatility, next, are features none of these one-factor models produces

Background: Fractional Brownian Motion and the Hurst Exponent (deeper)

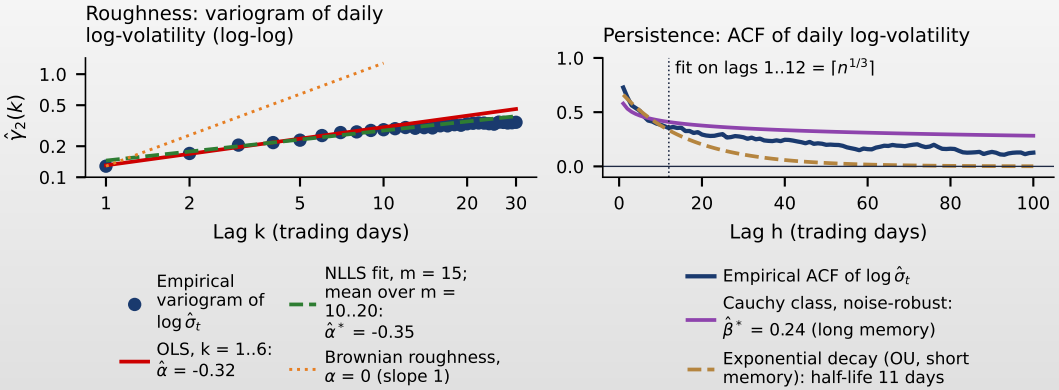
- **Fractional Brownian motion** B^H, $0 < H < 1$: a Gaussian process with $B_0^H = 0$ and $\mathrm{Cov}(B_s^H, B_t^H) = \frac{1}{2}\left(s^{2H} + t^{2H} - |t-s|^{2H}\right)$
  - $H = 1/2$ gives $\min(s,t)$, i.e. Brownian motion; increments have variance $\mathbb{E}[(B_{t+h}^H - B_t^H)^2] = h^{2H}$
  - $H < 1/2$: successive increments are negatively correlated and paths are rougher than Brownian motion; $H > 1/2$: positively correlated, smoother
- **Fractional OU**: the OU equation driven by B^H instead of W; one H sets both ends of its ACF
  - Short lags: the variogram behaves like $h^{2H} = h^{2\alpha+1}$, hence $\alpha = H - 1/2$
  - Long lags: the ACF decays like $h^{2H-2} = h^{-\beta}$, hence $\beta = 2 - 2H$
- Check: which H gives $\beta = 0.18$? Is that process rough?
  - Answer: $H = 1 - \beta/2 = 0.91$, so $\alpha = 0.41 > 0$: smooth, not rough; one H cannot match both features, which motivates the case study

Research Question: Rough, Yet Persistent?

- Paper: Bennedsen, Lunde & Pakkanen (2022), *Journal of Financial Econometrics*
  - Builds on Gatheral, Jaisson & Rosenbaum (2018): log-volatility as a fractional OU process with Hurst exponent $H < 1/2$ (RFSV, rough fractional stochastic volatility)
  - The long-memory model of Comte & Renault (1998) uses $H > 1/2$
- Roughness index α: $1 - \rho(h) \sim c|h|^{2\alpha+1}$ as $h \to 0$, with ρ the autocorrelation of log-volatility; $\alpha < 0$: paths rougher than Brownian motion
  - Memory parameter β: $\rho(h) \sim c|h|^{-\beta}$ as $h \to \infty$; $\beta < 1$: the ACF is not integrable, i.e. long memory
- Research questions (model, estimators and forecast results: ▸ Appendix)
  - are realised volatilities both rough and persistent?
  - do models that set α and β separately forecast better?

Case Study: Replication on SPY, $\Delta = 1$ Day (1/2)

NLLS: Non-Linear Least Squares.



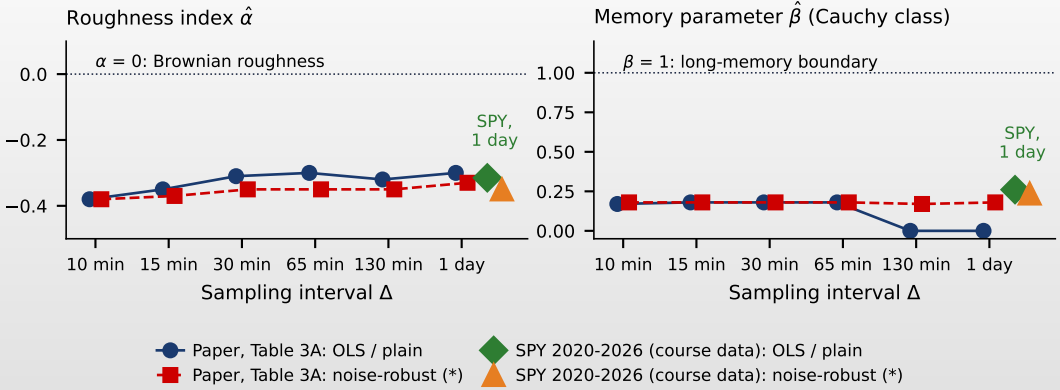
MFM_ch11_rough_vol

Case Study: Replication on SPY, $\Delta = 1$ Day (2/2)

- SPY 5-minute bars, 1 468 full days (October 2020 – September 2026); $x_t = \log \hat{\sigma}_t = \frac{1}{2} \log \mathrm{BV}_t$, daily bipower variation of the 78 five-minute returns Barndorff-Nielsen & Shephard (2004)
- Left, rough: variogram $\hat{\gamma}_2(k) = \frac{1}{n-k} \sum_t (x_{t+k} - x_t)^2$; $\hat{\alpha}_{\mathrm{OLS}} = -0.32$ ($k = 1, \ldots, 6$), noise-robust $\hat{\alpha}^*_{\mathrm{NLLS}} = -0.35$; Bennedsen, Lunde & Pakkanen (2022) at 1 day: -0.30, -0.33
- Right, persistent: ACF 0.13 at lag 100; Cauchy $\hat{\beta} = 0.26$, $\hat{\beta}^* = 0.24$ on 12 lags, below 1: consistent with long memory, not a proof; an OU fit (half-life 11 days) implies 0.002 at lag 100
- In the plot, m is the number of lags used in NLLS: 15 for the illustrated fit; the reported estimate averages fits over 10–20 lags.

MFM_ch11_rough_vol

Case Study: SPY Against Table 3 of the Paper (1/2)



MFM_ch11_rough_vol

Case Study: SPY Against Table 3 of the Paper (2/2)

- Lines: E-mini S&P 500 futures, 2011–2014, Δ from 10 minutes to 1 day (Bennedsen, Lunde & Pakkanen (2022), Table 3, Panel A); markers: SPY at 1 day, 2020–2026, course data; * = noise-robust
- $\hat{\alpha}$ barely moves with Δ (-0.38 to -0.30): roughness is a property of the process, and SPY falls in the same band; $\hat{\beta}^* = 0.17$–0.18 at every Δ, far below 1, while the plain $\hat{\beta}$ is unstable
- Limits: another contract, period and estimator (bars, no pre-averaging); realised measures estimate integrated, not spot, variance

MFM_ch11_rough_vol

An Open Question: How Much of the Price Path Is Discontinuous?

- Model: $\mathrm{d}X_t = \mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t + J_t\,\mathrm{d}N_t$ (X_t: log price; N_t: number of jumps, intensity λ; J_t: jump size); the answer depends on the sampling frequency and on the model
  - Daily Merton MLE (this chapter): jumps carry 71% of S&P 500 variance; the 5-minute bipower test (Chapter 9): 2.8% for SPY
  - Christensen, Oomen & Podolskij (2014): at tick frequency most jumps detected at coarser sampling disappear; they are rapid but continuous moves
  - Christensen, Oomen & Renò (2022): **drift bursts**, short explosive trends in μ_t, look like jumps in coarse data
- Formal question: H_0: $\lambda = 0$ (continuous path with stochastic volatility) against H_1: $\lambda > 0$
  - Non-standard: the boundary problem of the Merton LR test (Section 9)

The Discovery Loop and What the Researcher Checks

The loop with an AI (Artificial Intelligence) assistant

- Assistant: an LLM (Large Language Model) such as Claude, ChatGPT or Gemini
- **1. Literature**: published jump shares by frequency; check every Digital Object Identifier (DOI) on Crossref
- **2. Hypotheses**: Merton jumps, Heston volatility, drift bursts, each with a distinct testable implication
- **3. Replicate first** the chapter's estimates
- **4. Robustness**: filters and tests fixed before the first run
- **5. Critique**: the assistant challenges the inference, as a referee would
- **6. Report**: every specification, the code, the AI log
- Reusable prompts: Appendix

Checks: failure and test

- Boundary, unidentified parameters: parametric bootstrap under H_0
- Look-ahead in the volatility filter: standardise only with $\mathcal{F}_{t-1}$
- Frequency mismatch: jump shares at every frequency on a common sample
- Multimodal likelihood: many starting points and a profile in λ
- Invented numbers: every claim needs a table and a DOI

Mini-Case: Jumps or Stochastic Volatility in Disguise?

- A typical assistant proposal: "71% of variance comes from jumps"
  - Rival hypothesis: the Poisson mixture absorbs time-varying volatility; then standardising by EWMA volatility (Exponentially Weighted Moving Average; $\lambda_{\mathrm{EWMA}} = 0.94$, known at $t-1$) removes the jumps

	Raw returns	EWMA-standardised
Excess kurtosis	10.9	2.9
LR, Merton vs the Normal distribution	2 401	619
$\hat{\lambda}$ per year (s.e.)	102 (11)	124 (53)
Jump share of variance	71%	51%

- Standardising removes most kurtosis, yet the model still reports a "jump" about every other day: $\hat{\lambda}$ measures fat tails, not discontinuities
- Interpretation: daily data do not identify the jump share

Your Turn: A Project Seed

- Questions
  - how much of S&P 500 and Bitcoin variance is discontinuous?
  - how does this share depend on the sampling frequency?
  - are the detected "jumps" drift bursts?
- Replicate first: Merton jump share 71% (daily S&P 500), Lee–Mykland 0.87 jumps per year, bipower jump share 2.8% (SPY, 5 minutes, Chapter 9)
- Data: SPY 5-minute bars (October 2020 – September 2026) for every intraday frequency; S&P 500 and Bitcoin daily closes. The course data contain no intraday Bitcoin series: an intraday Bitcoin comparison needs an external source documented in the pre-registration

Your Turn: Deliverables and Event Window

- A pre-registration (hypotheses, frequencies, filters, tests, bootstrap design) dated before the first run
- Core deliverables, feasible with the course data
  - SPY: bipower jump share and Lee–Mykland counts at 5, 15 and 30 minutes; S&P 500 and Bitcoin: the daily values of this chapter
  - The EWMA standardisation of the mini-case, repeated on SPY at each frequency
  - The drift-burst statistic of Christensen, Oomen & Renò (2022) on SPY over 2–11 April 2025 (US tariff announcement on 2 April, 90-day pause on 9 April), against a calm window of equal length fixed in advance
- Optional extension: a bootstrap LR test of Heston (null, no jumps) against Bates
  - Both likelihoods need filtering of the latent variance, e.g. the Markov Chain Monte Carlo (MCMC) approach of Eraker, Johannes & Polson (2003); plan the computing time before committing to it
- Code, a two-page report with every specification tried, and a referee-style critique of your own result
- Declare your AI use in `AI_USE.md` and the errors you caught in `AI_ERRORS.md`: github.com/danpele/MFM/tree/main/project

Key Takeaways and Next: Chapter 12

- Under the assumptions of the Donsker theorem, rescaled random walks converge to Brownian motion; its quadratic variation t drives Itô calculus
- Itô's lemma gives exact solutions (GBM, Ornstein–Uhlenbeck, Merton); simulate exactly, or with Euler–Maruyama and Milstein at a known error order
- Identification: infill pins down volatility; drift and mean-reversion speed need long spans, and $\hat{\kappa}$ is biased upwards
- Prices are discounted Q-expectations because of no arbitrage; for continuous processes Girsanov changes only the drift; Feynman–Kac turns a Q-expectation into a PDE; $(\mathrm{VIX}/100)^2$ is a Q-expectation, affine in v_t
- In the fitted models jumps help with the tails and stochastic volatility with the clustering; volatility looks rough at short lags and persistent at long lags, which a one-factor OU model does not reproduce
- Next, Chapter 12: pricing options by replication and reading the volatility surface; code and data: github.com/danpele/MFM/tree/main/Quantlets/Ch_11

Check Your Understanding

- Why does a change of variable need an extra term in Itô calculus? Identify the role of quadratic variation.
- Does observing prices more frequently make the drift estimate precise? Separate frequency from the length of the observation period.
- Which feature calls for jumps, and which calls for stochastic volatility? Use the comparison with observed returns.
- Why can a pricing probability differ from a real-world probability? State what changes under the change of measure.

From Brown to Bachelier, Einstein and Wiener



Louis Bachelier (1870–1946)

Photo: unknown author; public domain; Wikimedia Commons



Albert Einstein in 1905

Photo: Lucien Chavan / ETH Zürich (1905); public domain; Wikimedia Commons

- 1827: the botanist Robert Brown sees pollen particles move in an irregular zig-zag under the microscope
- Bachelier (1900): French government bond (rente) prices and their options follow a random walk in continuous time
  - The spread of the price grows with $\sqrt{t}$, five years before physics
- Einstein (1905): molecular collisions explain Brown's motion; the mean squared displacement is linear in t
- Wiener (1923): Brownian motion as a probability measure on continuous paths, the **Wiener process**

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Itô, Samuelson, Black, Scholes and Merton



Kiyosi Itô (1915–2008)

Photo: Konrad Jacobs (1970); CC BY-SA 2.0 de; Wikimedia Commons



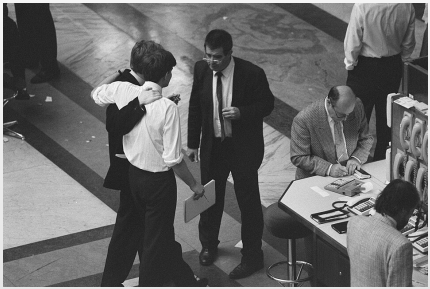
Paul A. Samuelson (1915–2009)

Photo: Bernard Gotfryd (1970–1975); public domain; Wikimedia Commons

- Itô (1944): the stochastic integral and its chain rule, the main working tool of this chapter
- Osborne (1959): log prices, not prices, behave like Brownian motion; Samuelson (1965): geometric Brownian motion
- Black & Scholes (1973) and Merton (1973): an option can be replicated in continuous time
- Merton (1976) adds jumps, Heston (1993) random volatility; textbook treatment: Franke, Härdle & Hafner (2019), Part I

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A Crash in Pictures, Amsterdam 1987



Amsterdam Stock Exchange, 21 October 1987, after the crash of 19 October

Photo: Bart Molendijk / Anefo (1987); CC0; Wikimedia Commons

- The fall of 19 October 1987 (Section 9): 22.6% in one day
- Under a Normal distribution such a day would not be expected once in the age of the universe: prices gap, and continuous models need jumps or fat-tailed innovations

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Data Cleaning and Technical Conditions

- Data: equity indices on weekdays, stale quotes removed (unchanged close and no intraday range), genuine zero returns kept; Bitcoin 7 days a week
- Itô integral: left-point sums converge for regular integrands (e.g. continuous and adapted); a general H is reached by approximation with simple processes
  - If only $\int_0^T H_t^2 \, \mathrm{d}t < \infty$ almost surely, $\int H \, \mathrm{d}W$ is merely a local martingale and its mean need not be zero
- SDE: existence and uniqueness of a strong solution if a and b are Lipschitz in x with at most linear growth Kloeden & Platen (1992)
  - Square-root coefficients (CIR, Heston) are not Lipschitz at zero, yet have unique strong solutions (Yamada–Watanabe conditions)
- Donsker: convergence in distribution in the space of càdlàg paths with the Skorokhod topology Billingsley (1999)

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Short Rates, More Numbers

- DTB3, 1954–2026: the rate ranged from -0.05% (2020-03-26) to 17.1% (1980-12-11); 4.3% of the days were below 0.05%
  - The fitted Vasicek model gives a 6.1% long-run probability of a negative rate
  - 2008–2026: 13 exact zeros and 8 negative quotes, outside the domain of the CIR and CKLS likelihoods; 17% of the days form a near-zero regime below 0.05%
- CKLS tests on γ, 1954–2007, HAC Wald with 11 (daily) and 6 (monthly) lags
  - $\gamma = 0$ rejected: daily 545, monthly 16; $\gamma = 1/2$ rejected daily (67.0; 22.1 with 55 lags), not monthly (0.5 against 3.84)
  - The Gaussian LR for $\gamma = 1/2$ rejects monthly (10.1): it ignores volatility clustering; Chan, Karolyi, Longstaff & Sanders (1992) found $\hat{\gamma} \approx 1.5$ for 1964–1989

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Vasicek Yield and the Observed Rate

- Vasicek bond with the Section 6 estimates, $\lambda = 0$ and $r = 3.99\%$: 10-year yield 4.05%; the 10-year Treasury yield (FRED series DGS10) on 2026-09-18: 5.01%
  - Conventions first: DGS10 is a semiannual par yield, $2\ln(1 + 5.01\%/2) = 4.95\%$ continuously compounded; DTB3 is a 91-day discount rate, 3.99% is 4.07% continuously compounded and lifts the model yield to 4.10%
  - The gap shrinks only from 0.96 to 0.85 percentage points (par versus zero-coupon yield is not adjusted): it is not a quoting artefact, but a term premium ($\lambda \neq 0$; matching it needs $\theta^Q > \hat{\theta}$, i.e. $\lambda < 0$) plus misspecification
- Nonparametric estimation: Stanton (1997) also estimates the market price of interest-rate risk; specification test: Aït-Sahalia (1996)

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The VIX Correction by κ^Q and the Full Model Comparison

κ^Q	1	2.5	5	10
b	0.96	0.90	0.82	0.68
Local correction $\hat{\xi}/b$	0.58	0.62	0.68	0.82

	Data	GBM	Merton	Heston
Excess kurtosis	10.88	0.00	3.66	3.30
Skewness	−0.36	0.00	−0.32	0.00†
Hill statistic	2.56	4.64	2.70	2.95
ACF of $\lvert r\rvert$, lag 1	0.27	0.00	0.00	0.28
ACF of $\lvert r\rvert$, sum of lags 1–20	5.32	0.00	0.00	4.78

- $\hat{\xi}/b$ is exact only at $v_t = \theta^Q$. Model columns: medians of 100 simulations
- A Heston model with a lower Feller ratio could reach the kurtosis: short-horizon excess kurtosis $\approx 3\xi^2/(2\kappa\theta)$
- † Zero by construction with the one-day Euler step (Section 11)

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Further Study: Reading

- Heavier tails: Lévy processes and α-stable distributions Cont & Tankov (2004)
- Stochastic calculus: Shreve (2004), Ch. 5–6; Franke, Härdle & Hafner (2019), chapters on the Wiener process and stochastic integration
- Exercises with solutions on all models of this chapter: Borak, Härdle & López-Cabrera (2013); the Heston model: Čížek, Härdle & Weron (2011); applications: Härdle, Chen & Overbeck (2017)
- Monte Carlo methods: Glasserman (2003); numerical SDEs: Kloeden & Platen (1992)

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Case Study, Model, Data and Method (1/2)

- **Model**: $\sigma_t = \xi e^{X_t}$ (eq. (1.3)), $X_t = \int_{-\infty}^{t} g(t-s)\, v_s \, \mathrm{d}W_s$, a Brownian semistationary (BSS) process with kernel g and volatility-of-volatility process v_s (eq. (1.5)); the forecasts take $v_s \equiv 1$, a Gaussian X (Section 5)
  - Gamma kernel $g(x) = x^{\alpha} e^{-\lambda x}$ (Example 1.2): roughness α, short memory with exponential decay at rate λ
  - Power kernel $g(x) = x^{\alpha}(1+x)^{-\gamma-\alpha}$ (eq. (1.8)): $\beta = \gamma$ if $\gamma > 1$, $\beta = 2\gamma - 1$ if $\gamma \in (1/2, 1)$ (Remark 1.2); a BSS process with this kernel is the Power-BSS model
- Gaussian alternative: the Cauchy class (Section 1.1.1)
  - $\rho(h) = (1 + |h|^{2\alpha+1})^{-\beta/(2\alpha+1)}$: roughness α and memory β set separately

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Case Study, Model, Data and Method (2/2)

- **Step 1**, α from the variogram $\hat{\gamma}_{\mathbf{2}}(k)$ of $\log \hat{\sigma}$: OLS of $\log \hat{\gamma}_{\mathbf{2}}(k)$ on $\log k$, $k = 1, \ldots, 6$, $\hat{\alpha} = (\hat{a}_{\mathbf{1}} - 1)/2$ (eq. (2.1))
  - Noise-robust NLLS: $\hat{\gamma}_{\mathbf{2}}(k) = b_{\mathbf{0}} + b_{\mathbf{1}}(k\Delta)^{\mathbf{2}\alpha+\mathbf{1}}$, the constant absorbs the measurement noise; average the estimates obtained with $m = 10, \ldots, 20$ lags (Sections 2.1 and 3.1)
  - **Step 2**: β, γ or λ by matching the model ACF to the empirical one on the first $\lceil n^{\mathbf{1}/\mathbf{3}} \rceil$ lags (Sections 2.2 and 4.1)
- **Data**: E-mini S&P 500 futures, every trade, 2011–2014 (996 days), 09:30–16:00 New York time; $\hat{\sigma}_t^{\mathbf{2}}$ = pre-averaged bipower variation over Δ, divided by Δ, for Δ from 10 minutes to 1 day (Sections 4 and 4.1)
  - Intraday seasonality removed with the flexible Fourier form of Andersen & Bollerslev (1997)
  - Daily data on 1 944 liquid US stocks, 2003–2013 (Sections 4.1 and 4.2)
- **Forecasts**: variance summed over the next $h = 1, 2, 5, 10, 20$ steps, rolling window of 200 observations, out of sample 2 January 2013 – 31 December 2014 (Section 5.1.1)
  - Rivals: random walk, AR(5), AR(10), ARMA(1,1), log-HAR (heterogeneous autoregressive model for log volatility) Corsi (2009), ARFIMA(0,d,0), RFSV
  - Losses MSE and QLIKE; 90% and 75% model confidence sets (MCS) Hansen, Lunde & Nason (2011); all defined on the next slide

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Case Study, Estimators and Forecast Losses

- **Variogram** $\hat{\gamma}_2(k)$ of $x_t = \log \hat{\sigma}_t$ (defined with the SPY replication in the main text); for small k, $\gamma_2(k) \approx c\, k^{2\alpha+1}$ (Section 2.1)
  - **Bipower variation** of the M returns r_j in a bin: $\mathrm{BV} = \frac{\pi}{2} \sum_{j=2}^{M} |r_j|\,|r_{j-1}|$, robust to jumps Barndorff-Nielsen & Shephard (2004)
  - Some texts, and Seminar 11, C3, scale BV by $M/(M-1)$: this shifts x_t by the constant $\frac{1}{2}\ln(78/77) \approx 0.006$ for $M = 78$, which cancels in $\hat{\gamma}_2(k)$ and in the ACF
  - Pre-averaging: returns are first averaged over short windows to damp microstructure noise (Appendix A of the paper)
- **Forecast object**: $\mathrm{FO}_t(\Delta, h) = \sum_{k=1}^{h} \mathrm{BV}_{t+k\Delta}$ (eq. (5.3)); for a Gaussian log model, $\mathbb{E}[e^{X_{t+\Delta}} \mid \mathcal{F}_t] = \exp(m + s^2/2)$, with m, s^2 the conditional mean and variance (eq. (5.2))
  - Losses over the out-of-sample origins: $\mathrm{MSE} = \mathrm{mean}\,(\widehat{\mathrm{FO}}_t - \mathrm{FO}_t)^2$, $\mathrm{QLIKE} = \mathrm{mean}\,(\log \widehat{\mathrm{FO}}_t + \mathrm{FO}_t/\widehat{\mathrm{FO}}_t)$ (Section 5.1.1)
- **Model confidence set** (MCS) Hansen, Lunde & Nason (2011): a set of models that contains the best model with probability 90% (or 75%)
  - Membership is not a ranking: the models inside the MCS cannot be told apart at that level

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Case Study, Results in the Paper

$\log\sigma$, E-mini S&P 500 (Table 3, Panel A)	$\hat{\alpha}_{\mathrm{OLS}}$	$\hat{\alpha}^*_{\mathrm{NLLS}}$	$\hat{\beta}^*_{\mathrm{Cauchy}}$	$\hat{\lambda}^*_{\mathrm{Gamma}}$
$\Delta = 10$ minutes	-0.38	-0.38	0.18	0.02
$\Delta = 65$ minutes	-0.30	-0.35	0.18	0.02
$\Delta = 1$ day	-0.30	-0.33	0.18	0.02

- Rough at every scale ($\hat{\alpha} < 0$, far rougher than Brownian motion, $\alpha = 0$) and persistent ($\hat{\beta}^* < 1$): long memory together with roughness; * = noise-robust; SPY comparison in the main text
  - 1 944 US stocks: $\hat{\alpha}^*$ mostly between -0.45 and -0.30, $\hat{\beta}^*$ mostly between 0 and 0.4, no systematic difference across sectors (Figures 4 and 5): both features look universal
- Noise can fake roughness: a simulated $\alpha = 0$ observed with noise-to-signal ratio 0.3 at $\Delta = 0.1$ gives a mean OLS estimate of -0.47 but a mean NLLS estimate of 0.01 (Table 1, Panel C)
- Forecasts: modelling persistence pays intraday and at long horizons
  - 15-minute steps, $h = 20$: MSE ($\times 10^{\mathbf{8}}$) 0.038 for log-HAR, 0.020 for RFSV, 0.019 for Power-BSS (Table 4, Panel A)
  - Daily steps, $h = 1$: RFSV is best, 0.493 ($\times 10^{\mathbf{9}}$) (Table 5, Panel B)

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Case Study, Intraday Application, Limits and Open Questions (1/2)

- **Intraday application** (illustrative, not an exact replication) on the SPY 5-minute bars of the course data
  - $\hat{\sigma}_t^2$ = bipower variation of the 5-minute returns in each bin of $\Delta = 30$ or 65 minutes
  - Divided by Δ and by the flexible Fourier seasonal factor Andersen & Bollerslev (1997)
  - OLS and NLLS $\hat{\alpha}$, Cauchy $\hat{\beta}^*$ and Gamma-BSS $\hat{\lambda}^*$ against Table 3, Panel A; log-HAR, RFSV and BSS forecasts, MSE, QLIKE and MCS against Table 5
  - Data limit: bars instead of every trade, so the pre-averaged estimator (Appendix A of the paper) is not feasible and the noise is larger; only 13 returns per 65-minute bin

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Case Study, Intraday Application, Limits and Open Questions (2/2)

- **Limits stated by the authors**
  - Realised measures are noisy estimates of integrated variance: the conclusions concern them, not the latent spot volatility (Sections 4.1, 4.2 and 6)
  - No rule for the NLLS bandwidth m (Sections 2.1 and 3.1); NLLS is too costly for rolling forecasts, which use the OLS $\hat{\alpha}$ (Section 5.1.1)
  - Forecasts assume a Gaussian X; a non-Gaussian volatility of volatility is left out of scope (Section 5; footnote 3)
- **Open question**: why does long memory improve even short-horizon forecasts? The authors leave it to further research (Section 5.1.2)
  - Rolling $\hat{\alpha}$ (65 minutes, 2005–2014, median -0.36) is less rough in 2007–2008 and 2011 (Figure 3)
  - $\hat{\alpha}$ rises with log volume, slope 0.0091 (Figure 6)
  - Does Bitcoin (5-minute bars), a market that never closes, fit this pattern?

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The Discovery Loop, Reusable Prompts (1/2)

- Tools: an AI assistant (an LLM such as Claude, ChatGPT, Gemini or GitHub Copilot); literature search: Semantic Scholar, Elicit, Google Scholar
- **1. Literature**: tabulate estimates by frequency, then check every DOI on Crossref (the title must match)
  - `Tabulate published estimates of the jump share of equity-index variance by sampling frequency (daily, 5-minute, tick) and method; give the exact number, the table it comes from, and the DOI.`
- **2. Competing hypotheses**, with distinct testable implications
  - `For three processes that give a daily excess kurtosis near 11 (Merton jumps, Heston volatility, drift bursts), derive the implications for returns standardised by ex-ante volatility, for bipower versus realised variance, and for the drift just before extreme moves.`
- **3. Data and code**: replicate the chapter's estimate first
  - `Write Python that reads the S&P 500 daily closes of the course and fits the Merton model by maximum likelihood (Poisson mixture truncated at 12 jumps, several starting points).`

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The Discovery Loop, Reusable Prompts (2/2)

- **4. Robustness**, pre-registered
  - `Write a pre-registered plan: standardise returns by EWMA and by GARCH volatility known at t-1; split 1990-2007 and 2008-2026; test Heston (no jumps, the null) against Bates (SVJ); obtain LR critical values by parametric bootstrap under the Heston null.`
- **5. Critique**: ask the assistant to attack the inference
  - `Act as a referee: why is the LR test of lambda = 0 non-standard, what is the correct reference distribution, and could a Poisson mixture with frequent small jumps simply be a fat-tailed innovation?`
- **6. Report**: all frequencies and filters, bootstrap critical values, code and the AI log

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Further Study on Quantinar

Advanced courses on quantinar.com

- Applied Computational Finance
- XFG Advanced Methods in Quantitative Finance

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_11

Acronyms Used in This Material (1/2)

- **ACF**: AutoCorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive model
- **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BET**: Bucharest Exchange Trading (index)
- **BSS**: Brownian SemiStationary (process)
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CBOE**: Chicago Board Options Exchange
- **CI**: Confidence Interval
- **CIR**: Cox–Ingersoll–Ross (model)
- **CKLS**: Chan–Karolyi–Longstaff–Sanders (model)
- **CLT**: Central Limit Theorem
- **DGS10**: 10-Year Treasury Constant Maturity Rate (FRED series code)
- **DTB3**: 3-Month Treasury Bill: secondary market rate (FRED series code)
- **FRED**: Federal Reserve Economic Data
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity

Acronyms Used in This Material (2/2)

- **GBM**: Geometric Brownian Motion
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **JB**: Jarque–Bera (test)
- **LLM**: Large Language Model
- **LR**: Likelihood Ratio (test)
- **MCS**: Model Confidence Set
- **MLE**: Maximum Likelihood Estimation
- **MSE**: Mean Squared Error
- **NLLS**: Non-Linear Least Squares
- **OLS**: Ordinary Least Squares
- **OU**: Ornstein–Uhlenbeck (process)
- **PDE**: Partial Differential Equation
- **QLIKE**: Quasi-LIKElihood (loss function)
- **QQ**: Quantile–Quantile (plot)
- **RFSV**: Rough Fractional Stochastic Volatility (model)
- **SDE**: Stochastic Differential Equation
- **SVJ**: Stochastic Volatility with Jumps (model)
- **US**: United States
- **VIX**: Cboe Volatility Index
- **XFG**: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

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